UTH UNIVERSITY OF TRANSPORT HOCHIMINH CITY

Trường đại học Giao Thông Vận Tải Hồ Chí Minh

Báo cáo

Hotel Room Booking System

Course code: 012012210505

Course name: Công nghệ phần mềm

Team Member:

Trần Ngọc Toàn Thắng - 066206017687

Cao Thị Kim Thư - 089307019882

Huỳnh Long Phi - 079207026464

Ngô Hoàng Ân - 079207039170

Nguyễn Duy Thuận - 075207008547

Nguyễn Đỗ Lê Khoa - 051207010823

Hồ Chí Minh, ngày 1 tháng 11 năm 2026

student ID number	Full name	level of contribution
079207039170	Ngô Hoàng Ân	100%
066206017687	Trần Ngọc Toàn Thắng	100%
089307019882	Cao Thị Kim Thư	100%
075207008547	Nguyễn Duy Thuận	100%
051207010823	Nguyễn Đỗ Lê Khoa	100%
079207026464	Huỳnh Long Phi	100%

Hotel Room Booking System Specifications

Version 1.1

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I. Hotel Room Booking System

The hotel reservation system has the following basic features:

- Accounts and Personal Information
  - The system allows new users to register an account by entering their full name, email address, phone number, and password. During registration, the system verifies that the email address has not been used, the phone number is valid, and the password meets security requirements. After successful registration, the user becomes a customer of the system.
  - Users log in using their email address and password. If the information is correct, the system directs them to the interface appropriate for their role—either customer or administrator. If incorrect, the system displays an error message and records the number of failed login attempts. Once logged in, users can log out to end their session. If they forget their password, they enter their registered email address to receive password reset instructions via email. Users can also proactively change their password by entering their old password and a new one.
  - Once they have an account, customers can update their personal information, including full name, phone number, national ID number, address, gender, and date of birth.

- **View, Search, and Filter Rooms**
  - All users, including those who are not logged in, can view the list of rooms and prices. Each room displays actual photos, price per night, capacity, room type, and availability. When viewing a room's details, users see additional information such as a detailed description, square footage, included amenities, and reviews from previous guests.
  - Users can search for rooms by keyword, check-in date, check-out date, and number of guests. The system returns only rooms that are available during that date range and have sufficient capacity. If a user makes a spelling mistake, the system will still attempt to return relevant results. Results can be further filtered by price range and availability status.

- **Room Reservations and Reservation Management**
  - Only users with an account can select and book rooms. If a guest who is not logged in attempts to select a room, the system redirects them to the login page and then returns them to the step they were on.
  - When making a reservation, guests select a room, check-in date, check-out date, and enter the number of adults and children. The system checks room availability and creates a reservation request in "pending" status. A single reservation may include multiple rooms. During the booking process, guests can add additional services such as spa treatments or breakfast; the cost of these services will be added to the total cost.
  - Customers can review their reservations by status: pending, confirmed, in-stay, completed, and canceled. Customers can also cancel a reservation in accordance with the hotel's policy. The system checks the cancellation conditions before changing the reservation to "canceled" status and returning the room to "available" status. After

completing their stay, guests can rate the room with a star rating and write a review. This feature is only available once the reservation has been completed.

+ For guests arriving directly at the hotel without an account, staff create an account for the guest using the user management feature, then book a room for them as they would for any other guest.

- **Billing and Payment**
  + The system automatically calculates the total cost based on the number of nights stayed and the number of rooms, plus services, taxes, and surcharges (if any), then deducts any discounts. Customers can enter a discount code. The system verifies that the code exists, is valid, and is currently active; only one code can be used per booking.
  + Before payment, the system displays a summary including the base amount, the discount from the voucher, the deposit, taxes, and the total, and asks the customer to confirm. Customers can pay online via the payment gateway or choose to pay at the hotel. Upon successful payment, the system records the transaction, issues an invoice linked to the reservation, and sends a confirmation email to the customer.
  + If a guest cancels a booking in accordance with the policy and the booking was paid for online, the system initiates a refund through the payment gateway, records the refund amount, and updates the transaction status to "refunded." Bookings paid for at the hotel do not go through this step.

- **Administration**
  + An administrator has full authority to manage the system and, like all other users, can view, search for, and filter rooms.
  + Regarding data, administrators manage rooms and room types (add, edit, delete, update prices, images, and descriptions), manage services and assign them to rooms, manage discount codes, and manage users—including viewing the user list, locking or unlocking accounts, and creating accounts for walk-in guests.
  + Regarding reservations, administrators view all guest reservations and approve pending reservations to move them to "Confirmed" status. When guests arrive, administrators check in confirmed reservations, verify ID cards against reservation information, and move the reservation to "Staying." During the stay, any additional charges incurred are recorded on the reservation along with the reason. When guests check out, the administrator processes the checkout; the system finalizes the additional charges, recalculates the total amount, issues an invoice, marks the reservation as "completed," and returns the room to "vacant" status. Finally,

the administrator can view the hotel's revenue—a privilege reserved exclusively for administrators.

II. Supplementary Specification

Usability
The system assists users when they make spelling mistakes during a search, preventing situations where a search returns no results simply because of a few incorrect characters. Additionally, before checkout, the system must confirm the details with the customer and clearly and accurately display each item—including the base price, voucher amount, deposit, tax, and total—so the customer can be confident about the amount due.

Reliability
Once a room has been booked for a specific time period, the system must lock that room to prevent multiple people from booking it simultaneously. If two people attempt to book the same room at the same time, only one booking will be successful. Furthermore, the system must operate reliably 24/7 so that customers can book rooms at any time.

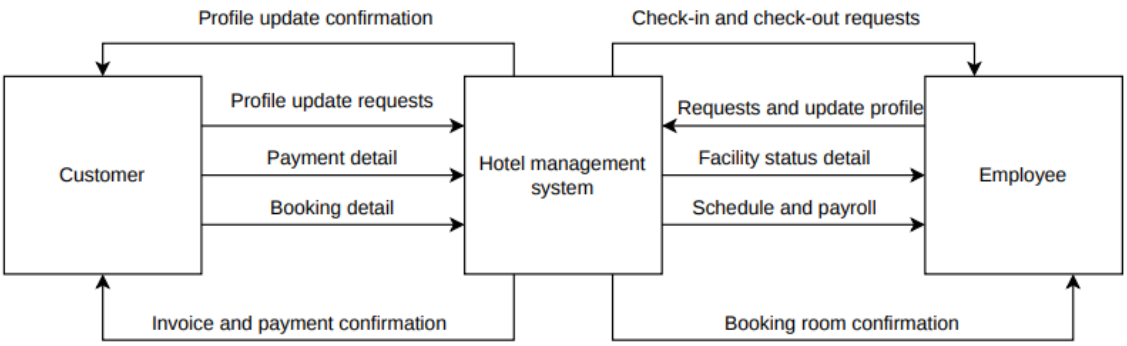
Performance
The system must be able to handle 5,000 concurrent users while continuing to function normally. Page reload time must not exceed 3 seconds. When users filter information based on their criteria, results must be returned within 3 seconds.

Security
Passwords must include lowercase letters, uppercase letters, numbers, and special characters. If an account enters the wrong password more than 3 times, the system sends a warning email to the account owner. The system also sends a notification email every time a password is changed. Access permissions are strictly enforced: only administrators are allowed to view revenue and use administrative functions; customers attempting to access these functions will be denied.

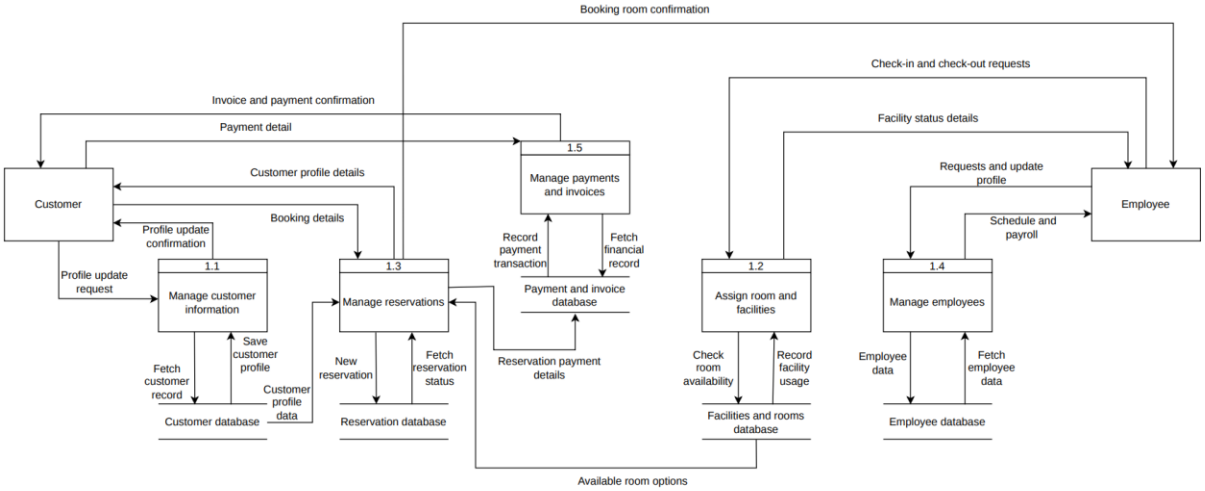
Development Requirements
Web-based applications. HTML + CSS + the Python programming language + the Figma UI design platform + the GitHub code hosting platform + the Draw.io diagramming platform + Google Docs text creation and editing tool

III. Data Flow Diagram (DFD)

Level 0:

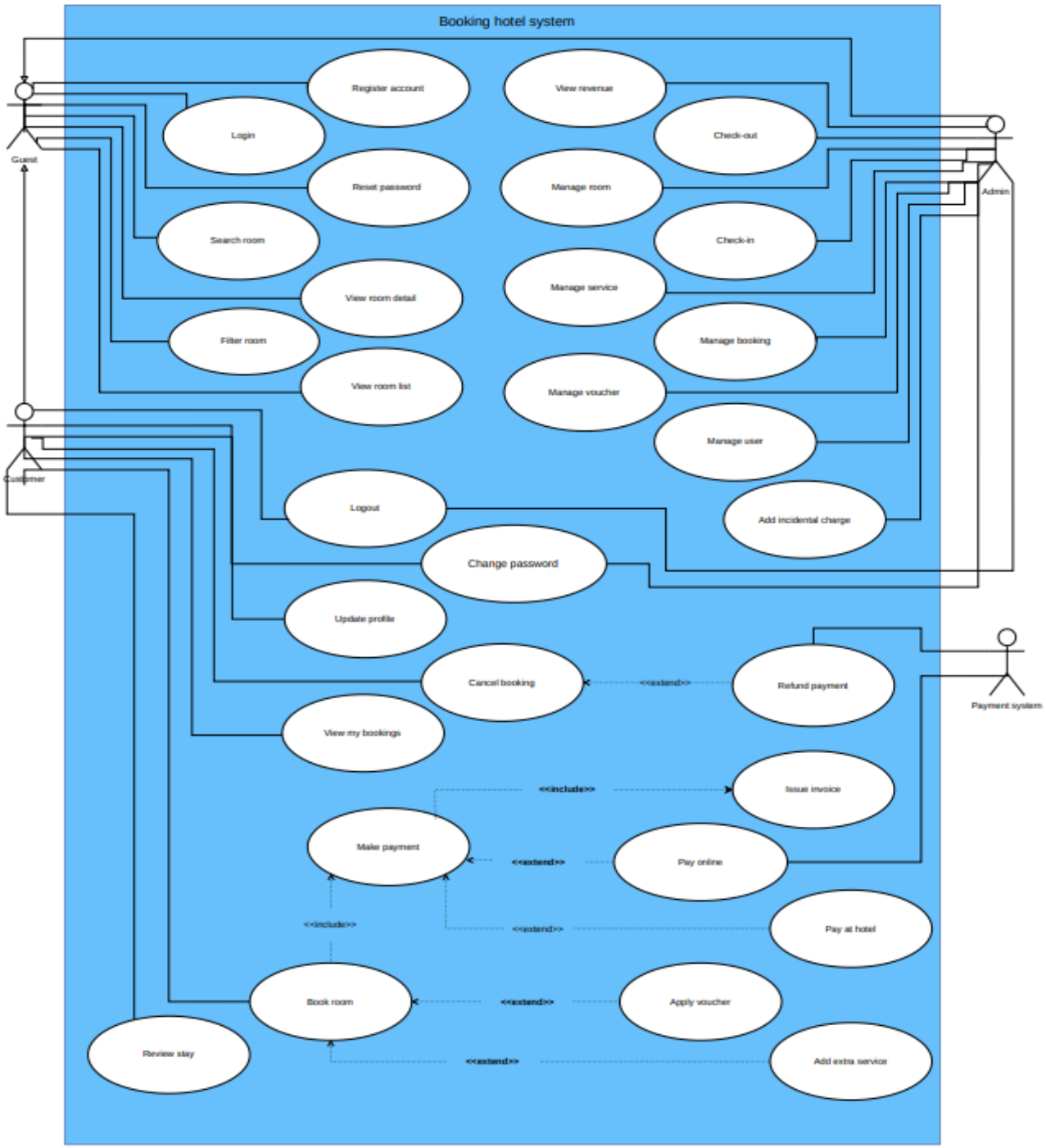


Level 1:



IV. Use-Case diagram

1. overview diagram



1.1. List of actors

No.	Actor	Description
1	Guest	A general system user, including those without an account. Can view, search, and filter rooms; register; and log in. The Customer and Admin actors inherit from this actor.
2	Customer	A customer with an account. In addition to the Guest's functions, can book rooms, make payments, cancel reservations, and leave reviews.
3	Admin	Administrator (hotel staff). In addition to the Guest's functions, they can manage data and reservations, handle check-ins and check-outs, and view revenue.
4	Payment system	External payment gateway. Processes online payment transactions and refunds.

1.2. List of Use-cases

No.	Use-cases	Description
1	Register account	Sign up for a new account using your full name, email address, phone number, and password.
2	Login	Log in to the system with your email and password
3	Reset password	Reset your password via email if you forget it.
4	View room list	View the room list with photos, nightly rates, capacity, and availability.
5	View room detail	View details about a room: description, size, included amenities, and reviews from previous guests.
6	Search room	Search for rooms by keyword, check-in date, check-out date, and number of guests.
7	Filter room	Filter rooms by price range and availability.
8	Logout	Log out, end your session.
9	Change password	Change your password by entering your old password and your new password.
10	Update profile	Update personal information: full name, phone number, national ID number, address, gender, date of birth.
11	Book room	Select a room, check-in date, check-out date, and number of guests, then create a reservation.
12	Apply voucher	Enter the discount code when booking a room (optional; click "Book Room" to expand).

13	Add extra service	Add additional services such as spa treatments and breakfast when booking a room (optional, under “Book a Room”).
14	Make payment	Pay for your reservation, including reviewing the cost summary and selecting a payment method.
15	Pay online	Pay online through the payment gateway.
16	Pay at hotel	Pay directly at the hotel.
17	Issue invoice	Issue an invoice and send a confirmation email after payment is successfully processed.
18	View my bookings	View your reservations by status.
19	Cancel booking	Cancel the reservation in accordance with the hotel's policy.
20	Refund payment	Refunds for valid cancellations and orders paid online
21	Review stay	Rate with stars and leave a review once your stay is complete.
22	Manage room	Add, edit, and delete rooms and room types.
23	Manage service	Add, edit, and delete services, and assign services to rooms.
24	Manage voucher	Create, edit, and disable discount codes.
25	Manage user	View the list of users, lock or unlock accounts, and create accounts for walk-in customers.
26	Manage booking	View all reservations, review reservations, and cancel reservations.
27	Add incidental charge	Record any additional charges on the order form along with the reason.
28	Check-in	Confirm the guest's check-in, verify their ID card, and update their status to “currently staying.”
29	Check-out	Confirm guest checkout, finalize additional charges, issue an invoice, and mark the reservation as completed.
30	View revenue	View hotel revenue; only Admins can access this.

2. Use-case Register account

2.1. Summary

This use case describes how a Guest registers a new account using full name, email, phone number and password so that they can book rooms.

2.2. Flow of events

2.2.1. Basic flow

The use case begins when the user selects the Register function.

1. The system asks the user to provide: full name, email, phone number, password and password confirmation.
2. The user enters the information and submits the registration request.
3. The system validates the information, encrypts the password, and adds the new account (role Customer) together with a customer profile to the system.
4. The system notifies the user that registration was successful and redirects the user to the login function.

2.2.2. Alternative flows

2.2.2.1. Email (username) already exists

If, in the Basic flow, the user enters an email that already exists in the system, the system displays an error message. The user may return to the beginning of the Basic flow or cancel the operation, in which case the use case ends.

2.2.2.2. Missing or invalid information

If the user leaves out required information (full name, email, phone number, password), enters an email or phone number in an invalid format, uses a password that violates the password rules, or the password confirmation does not match, the system displays an error message specifying the invalid information. The user may return to the beginning of the Basic flow or cancel the operation, in which case the use case ends.

2.3. Special requirements

- The password must have at least 8 characters, including lowercase letters, uppercase letters, digits and special characters.

2.4. Pre-conditions

None.

2.5. Post-conditions

If the use case succeeds, the new account and customer profile are added to the system. Otherwise, the system state is unchanged.

2.6. Extension points

None.

3. Use-case Login

3.1. Summary

This use case describes how a user (guest or Admin) logs in to the Hotel Booking System using email and password.

3.2. Flow of events

3.2.1. Basic flow

The use case begins when the user selects the Login function.

1. The system asks the user to enter email and password.
2. The user enters email and password and submits the login request.
3. The system verifies the email and password. If they are correct, the system resets the failed-attempt counter to 0 and allows the user to log in.
4. The system redirects the user to the page corresponding to their role: the home page for customers, the administration page for Admin.

3.2.2. Alternative flows

3.2.2.1. Incorrect email or password

If, in the Basic flow, the user enters an incorrect email or password, the system displays an error message and increases the consecutive failed-attempt count of the account (if the email exists). The user may return to the beginning of the Basic flow or cancel the operation, in which case the use case ends.

3.2.2.2. More than 3 failed password attempts

If the user enters an incorrect password more than 3 consecutive times, the system locks login for that account and sends a warning email to the account's email address. The account can log in again only after the user recovers the password or an Admin unlocks it. The use case ends.

3.2.2.3. Account is locked

If the account is locked (due to more than 3 failed attempts or locked by an Admin), the system displays a message that the account cannot log in and gives instructions to recover the password or contact the hotel. The use case ends.

3.3. Special requirements

- Login is locked after 3 failed password attempts to prevent password guessing.

3.4. Pre-conditions

The user is not logged in to the system.

3.5. Post-conditions

If the use case succeeds, the user is logged in to the system. Otherwise, the system state is unchanged (except that the consecutive failed-attempt count is recorded).

3.6. Extension points

- Use case Reset password: The user may switch to the password recovery function when they have forgotten their password.

4. Use-case Reset password

4.1. Summary

This use case allows a user to recover their password via email when they have forgotten it.

4.2. Flow of events

4.2.1. Basic flow

The use case begins when the user selects the Reset password function.

1. The system asks the user to enter their email.
2. The user enters their email and requests password recovery.
3. The system checks whether the email exists in the system. If it exists, the system generates a new random password, resets the failed-attempt counter to 0 (unlocking the account if it was locked due to failed attempts), and sends the new password to the user's email.
4. The system notifies the user of the result of the password recovery.

4.2.2. Alternative flows

4.2.2.1. Email does not exist in the system

If the email does not exist, the system notifies the user. The user may restart this function or exit it.

4.2.2.2. Failed to send email

If the system cannot send the email, the system displays an error message and keeps the old password unchanged. The user may try again later.

4.3. Special requirements

- The new password must be randomly generated and comply with the password rules (at least 8 characters, including lowercase letters, uppercase letters, digits and special characters).

4.4. Pre-conditions

None.

4.5. Post-conditions

If the use case succeeds, the password is changed and an email containing the new password is sent to the user. Otherwise, the system state is unchanged.

4.6. Extension points

None.

5. Use-case View room list

5.1. Summary

This use case allows a user to view the list of rooms with images, price per night, capacity and availability.

5.2. Flow of events

5.2.1. Basic flow

The use case begins when the user selects the function to view the room list.

1. The system displays the list of rooms; each room shows its image, room number, room type, price per night, capacity and availability. The list is paginated.
2. If the user selects another page, the system displays the list of rooms on that page.
3. If the user selects to view the details of a room, the system switches to the View room detail function.

5.2.2. Alternative flows

5.2.2.1. No rooms available

If the system has no rooms yet, the system displays a message that there are no rooms to display.

5.3. Special requirements

- For at least 95% of requests, the room list must be displayed within 3 seconds.
- All users, including those who are not logged in, can view the room list and prices.

5.4. Pre-conditions

None.

5.5. Post-conditions

None.

5.6. Extension points

- Use case View room detail: If the user chooses to view the details of a room.
- Use case Search room: If the user chooses to search for rooms.
- Use case Filter room: If the user chooses to filter the room list.

6. Use-case View room detail

6.1. Summary

This use case allows a user to view the detailed information of a room: description, area, amenities and included services, and reviews from guests who have stayed.

6.2. Flow of events

6.2.1. Basic flow

The use case begins when the user selects to view the details of a room.

1. The system displays the detailed information of the room, including: room number, room type, images, detailed description, area, capacity, price per night, included services and amenities, and the cancellation policy.
2. The system displays the reviews from guests who have stayed: the average rating and the list of reviews (number of stars, content, review date), with the newest review first.

6.2.2. Alternative flows

6.2.2.1. Room does not exist or is no longer in operation

If the room no longer exists or is no longer in operation, the system displays a message and returns the user to the room list.

6.2.2.2. Room has no reviews

If the room has no reviews yet, the system displays a message that there are no reviews.

6.3. Special requirements

- All users, including those who are not logged in, can view room details and prices.

6.4. Pre-conditions

None.

6.5. Post-conditions

None.

6.6. Extension points

- Use case Book room: If the user chooses to book the room being viewed.

7. Use-case Search room

7.1. Summary

This use case allows a user to search for rooms by keyword, check-in date, check-out date and number of guests.

7.2. Flow of events

7.2.1. Basic flow

The use case begins when the user selects the room search function.

1. The system displays a search form containing: keyword (room name or room type), check-in date, check-out date, and number of guests.
2. The user enters the criteria and submits the search request.
3. The system validates the criteria, then finds and lists the rooms that match the keyword, have enough capacity for the number of guests, and are available during the selected period (image, room number, room type, price per night, capacity). The list is paginated, with at most 10 rooms per page.
4. If the user selects another page, the system displays the list of rooms on that page.

7.2.2. Alternative flows

7.2.2.1. Invalid search criteria

If the check-in date is before the current date, the check-out date is not after the check-in date, or the number of guests is invalid, the system displays an error message and asks the user to re-enter the criteria.

7.2.2.2. No matching rooms

If no room satisfies the criteria, the system notifies the user that no rooms were found and suggests changing the criteria.

7.3. Special requirements

- For at least 95% of requests, the search results must be returned within 3 seconds.
- Only rooms that have no active booking overlapping the selected period are listed.

7.4. Pre-conditions

None.

7.5. Post-conditions

None.

7.6. Extension points

- Use case View room detail: If the user chooses to view the details of a room.

- Use case Filter room: If the user chooses to filter the search results.

8. Use-case Filter room

8.1. Summary

This use case allows a user to filter the room list by price range and room availability.

8.2. Flow of events

8.2.1. Basic flow

The use case begins when the user selects the room filter function on the room list or the search results.

1. The system displays a filter containing: minimum price, maximum price, and room status (available or all).
2. The user enters the filter criteria and requests to apply them.
3. The system validates the criteria and displays the rooms in the current list that satisfy the filter criteria, with pagination.
4. If the user chooses to clear the filter, the system displays the original room list again.

8.2.2. Alternative flows

8.2.2.1. Invalid price range

If the minimum price is greater than the maximum price or a price entered is negative, the system displays an error message and asks the user to re-enter.

8.2.2.2. No rooms match the filter

If no room satisfies the filter, the system notifies the user that no rooms were found and suggests changing the filter.

8.3. Special requirements

- For at least 95% of requests, the filter results must be returned within 3 seconds.

8.4. Pre-conditions

The user is viewing the room list or the room search results.

8.5. Post-conditions

None.

8.6. Extension points

None.

9. Use-case Logout

9.1. Summary

This use case allows a logged-in user to log out of the system and end their session.

9.2. Flow of events

9.2.1. Basic flow

The use case begins when the user selects the Logout function.

1. The system ends the user's login session.
2. The system redirects the user to the home page in the not-logged-in state.

9.2.2. Alternative flows

None.

9.3. Special requirements

None.

9.4. Pre-conditions

The user must be logged in to the system.

9.5. Post-conditions

If the use case succeeds, the login session ends and the user returns to the not-logged-in state. Otherwise, the system state is unchanged.

9.6. Extension points

None.

10. Use-case Change password

10.1. Summary

This use case allows a logged-in user to change their password by entering the old password and a new password.

10.2. Flow of events

10.2.1. Basic flow

The use case begins when the user selects the Change password function.

1. The system asks the user to enter the old password, the new password and the new password confirmation.
2. The user enters the information and requests the password change.

3. The system checks whether the old password is correct, whether the new password complies with the password rules and differs from the old password, and whether the confirmation matches. If valid, the system updates the new password.
4. The system sends an email notifying the user that the password was changed successfully.
5. The system notifies the user of the result of the password change.

10.2.2. Alternative flows

10.2.2.1. Invalid input

If the old password is incorrect, the new password does not comply with the rules or is the same as the old password, or the new password confirmation does not match, the system displays an error message. The user may return to the beginning of the Basic flow or cancel the operation, in which case the use case ends.

10.3. Special requirements

- The password must have at least 8 characters, including lowercase letters, uppercase letters, digits and special characters.

10.4. Pre-conditions

The user must be logged in to the system.

10.5. Post-conditions

If the use case succeeds, the password is updated and a notification email is sent to the user. Otherwise, the system state is unchanged.

10.6. Extension points

None.

11. Use-case Update profile

11.1. Summary

This use case allows a logged-in customer to update their personal information: full name, phone number, citizen ID number (CCCD), address, gender and date of birth.

11.2. Flow of events

11.2.1. Basic flow

The use case begins when the customer selects the function to edit their personal information.

1. The system displays the customer's current personal information: full name, email, phone number, citizen ID number (optional), address, gender and date of birth (the password is not displayed).
2. The customer edits the information as desired (except the email).
3. The customer clicks the update button.
4. The system validates the information and updates it in the system.
5. The system notifies the user of the result of the update.

11.2.2. Alternative flows

11.2.2.1. Customer cancels the operation

If the customer cancels the operation, the system does not update the information and the use case ends.

11.2.2.2. Invalid input

If the customer leaves required information blank (full name, phone number), the phone number has an invalid format, the citizen ID number has an invalid format (not 12 digits), or the date of birth is later than the current date, the system displays an error message. The user may return to the beginning of the Basic flow or cancel the operation, in which case the use case ends.

11.3. Special requirements

- A customer can only view and update their own information.
- The citizen ID number is sensitive personal data; it is protected and can be viewed only by the account owner and Admin. The customer may leave it blank and add it later (Admin will verify it at check-in).

11.4. Pre-conditions

The customer (Customer) must be logged in to the system.

11.5. Post-conditions

If the use case succeeds, the personal information is updated. Otherwise, the system state is unchanged.

11.6. Extension points

None.

12. Use-case Book room

12.1. Summary

This use case allows a customer to select a room, choose the check-in date, check-out date and number of guests, and then create a booking.

12.2. Flow of events

12.2.1. Basic flow

The use case begins when the customer selects “Book Room” for a room (from the room list, the search results or the room detail page).

1. The system displays the booking form containing room information (room number, room type, price per night, capacity), check-in date, check-out date, number of adults and number of children (defaulting to the search criteria if any).
2. The customer selects or edits the above information and submits the booking request.
3. The system validates the information (valid dates, number of guests not exceeding capacity) and checks that the room is available during the selected period.
4. If the customer chooses to enter a discount code, the system performs the Apply voucher function.
5. If the customer chooses to add extra services, the system performs the Add extra service function.
6. The system creates the booking with the status “Pending” (including room details, number of nights, room price and subtotal), temporarily holds the room for 15 minutes, and calculates the total cost (room charge, services, VAT, discount) and the deposit (30% of the total).
7. The system redirects the customer to the payment function (Make payment).

12.2.2. Alternative flows

12.2.2.1. Customer is not logged in

If the user is not logged in when choosing to book a room, the system asks the user to log in or register an account, and then returns to the booking step in progress.

12.2.2.2. Invalid input

If the check-in date is before the current date, the check-out date is not after the check-in date, or the number of guests exceeds the room capacity, the system displays an error message. The user may return to the beginning of the Basic flow or cancel the operation, in which case the use case ends.

12.2.2.3. Room is no longer available

If the room has been booked by someone else during the selected period, the system notifies the customer that the room is no longer available and takes the customer back to the room search function.

12.2.2.4. Room hold expired

If more than 15 minutes have passed since the temporary hold and the customer has not completed the payment step, the booking is automatically changed to “Cancelled” and the room is released. The customer must book again.

12.3. Special requirements

- A room cannot be booked for two overlapping periods, even when many customers book at the same time.
- The room is temporarily held for 15 minutes from the time the booking is created.
- The amounts displayed and the amounts recorded in the booking must be absolutely accurate.

12.4. Pre-conditions

The customer (Customer) must be logged in to the system.

12.5. Post-conditions

If the use case succeeds, the booking is added to the system with the status “Pending” and the room is held for the booked period. Otherwise, the system state is unchanged.

12.6. Extension points

- Use case Apply voucher: If the customer chooses to enter a discount code.
- Use case Add extra service: If the customer chooses to add extra services.

13. Use-case Apply voucher

13.1. Summary

This use case allows a customer to enter a discount code (optional) when booking in order to receive a discount.

13.2. Flow of events

13.2.1. Basic flow

The use case begins when the customer chooses to enter a discount code during the booking step.

1. The system asks the customer to enter the voucher code.

2. The customer enters the voucher code and requests to apply it.
3. The system validates the voucher code: the code exists, is active, has not expired, and has not already been applied to this booking.
4. If the code is valid, the system records the voucher for the booking, recalculates the discount amount, deposit, VAT and total, and displays them to the customer again.

13.2.2. Alternative flows

13.2.2.1. Invalid voucher code

If the code does not exist, has expired, is inactive, or has already been applied to this booking, the system displays an error message stating the reason. The customer may enter another code or skip applying a voucher, in which case the use case ends.

13.3. Special requirements

- A booking may have multiple vouchers applied; each voucher can be applied only once per booking; the total discount must not exceed the total amount of the booking.
- The amount after the discount must be calculated with absolute accuracy.

13.4. Pre-conditions

The customer must be logged in and in the process of booking a room.

13.5. Post-conditions

If the use case succeeds, the voucher is recorded for the booking and the total is updated. Otherwise, the system state is unchanged.

13.6. Extension points

None.

14. Use-case Add extra service

14.1. Summary

This use case allows a customer to add extra services such as spa and breakfast when booking (optional).

14.2. Flow of events

14.2.1. Basic flow

The use case begins when the customer chooses to add extra services during the booking step.

1. The system displays the list of services offered by the selected room, including service name and price.

2. The customer selects one or more services and requests to add them to the booking.
3. The system validates the services (offered by the room, not already in the booking), adds the services to the booking, recalculates the total cost and the deposit, and displays them to the customer again.
4. If the customer chooses to remove a service that was added (while the booking is unpaid), the system removes the service from the booking and recalculates the total cost.

14.2.2. Alternative flows

14.2.2.1. No service selected or invalid service

If the customer selects no service, or a service is no longer offered by the room, the system displays a message and leaves the booking unchanged. The customer may select again or skip adding services, in which case the use case ends.

14.3. Special requirements

- The service price is recorded as of the time it is added; later price changes do not affect existing bookings.
- The amount after adding services must be calculated with absolute accuracy.

14.4. Pre-conditions

The customer must be logged in and in the process of booking a room (the booking is in “Pending” status and unpaid).

14.5. Post-conditions

If the use case succeeds, the services are added to the booking and the total is updated. Otherwise, the system state is unchanged.

14.6. Extension points

None.

15. Use-case Make payment

15.1. Summary

This use case allows a customer to pay for a booking, including reviewing the cost summary and choosing a payment method.

15.2. Flow of events

15.2.1. Basic flow

The use case begins when the customer proceeds to the payment step after booking.

1. The system displays the cost summary of the booking, including: base price (room charge, services), voucher discount, VAT, deposit (30% of the total) and the grand total.
2. The customer reviews and confirms the booking.
3. The system asks the customer to choose a payment method: online payment (paying the deposit through a payment gateway) or payment at the hotel.
4. If the customer chooses online payment, the system performs the Pay online function.
5. If the customer chooses payment at the hotel, the system performs the Pay at hotel function.

15.2.2. Alternative flows

15.2.2.1. Room hold expired

If more than 15 minutes have passed since the booking was created and the customer has not finished choosing a payment method, the booking is automatically changed to "Cancelled", the room is released, and the system notifies the customer.

15.2.2.2. Customer does not confirm

If the customer does not confirm or leaves the payment page, the booking is kept for the remaining hold time; after that, the Room hold expired flow applies.

15.3. Special requirements

- The amounts must be displayed clearly and with absolute accuracy before the customer confirms.
- The deposit equals 30% of the total and is collected only when the customer chooses online payment.

15.4. Pre-conditions

The customer must be logged in, have created a booking in "Pending" status, and the room must be temporarily held.

15.5. Post-conditions

If the use case succeeds, the payment method of the booking is recorded. Otherwise, the system state is unchanged.

15.6. Extension points

- Use case Pay online: If the customer chooses online payment.
- Use case Pay at hotel: If the customer chooses payment at the hotel.

16. Use-case Pay online

16.1. Summary

This use case allows a customer to pay the deposit of a booking through an online payment gateway (Payment System).

16.2. Flow of events

16.2.1. Basic flow

The use case begins when the customer chooses the online payment method at the payment step.

1. The system creates a payment transaction for the deposit amount and redirects the customer to the online payment gateway.
2. The customer enters payment information and confirms on the payment gateway.
3. The payment gateway (Payment System) processes the transaction and returns the result to the system.
4. If the payment succeeds, the system records the payment (amount, method, time, successful status) for the booking.
5. The system performs the Issue invoice function and sends a confirmation email.
6. The system takes the customer to the booking result page; the booking remains in “Pending” status until the Admin confirms it.

16.2.2. Alternative flows

16.2.2.1. Payment failed or cancelled

If the payment fails, the customer cancels at the payment gateway, or the payment gateway does not respond, the system does not record any payment and displays an error message. The customer may try again, choose payment at the hotel, or stop the booking; when stopping, the booking is cancelled and the room is released.

16.2.2.2. Room hold expired

If the 15-minute room hold has expired, the system releases the room and does not allow payment to continue. If a successful payment result is returned after that time and the room is no longer available, the system does not keep the booking and refunds the money to the customer (use case Refund payment).

16.3. Special requirements

- The system does not store the customer's payment card information; entering and processing card information is handled entirely by the payment gateway.
- The amount sent to the payment gateway must be absolutely accurate.

16.4. Pre-conditions

The customer must be logged in, have chosen online payment, and the room must be temporarily held.

16.5. Post-conditions

If the use case succeeds, the deposit payment is recorded, an invoice is issued and a confirmation email is sent to the customer. Otherwise, the system state is unchanged.

16.6. Extension points

None.

17. Use-case Pay at hotel

17.1. Summary

This use case allows a customer to choose to pay directly at the hotel instead of paying online.

17.2. Flow of events

17.2.1. Basic flow

The use case begins when the customer chooses the pay-at-hotel method at the payment step.

1. The system informs the customer that no deposit will be collected when booking and that the entire cost will be paid directly at the hotel upon check-out.
2. The customer confirms the choice of payment at the hotel.
3. The system records the payment method “at hotel” for the booking (deposit equal to 0).
4. The system sends a booking confirmation email, including the booking code and the full booking information, to the customer's email.
5. The system takes the customer to the booking result page; the booking remains in “Pending” status until the Admin confirms it.

17.2.2. Alternative flows

17.2.2.1. Customer does not confirm

If the customer does not confirm, the customer may go back and choose online payment or stop the booking; when stopping, the booking is cancelled and the room is released.

17.2.2.2. Failed to send email

If the system cannot send the email, the booking is still recorded and the system notifies the customer to review the booking in the function for viewing placed bookings.

17.3. Special requirements

- The payment at the hotel is recorded when the Admin performs the check-out.

17.4. Pre-conditions

The customer must be logged in, have created a booking in “Pending” status, and the room must be temporarily held.

17.5. Post-conditions

If the use case succeeds, the pay-at-hotel method is recorded for the booking and a confirmation email is sent to the customer. Otherwise, the system state is unchanged.

17.6. Extension points

None.

18. Use-case Issue invoice

18.1. Summary

This use case issues an invoice for a booking and sends a confirmation email to the customer after a payment has been processed successfully.

18.2. Flow of events

18.2.1. Basic flow

The use case begins when a payment for the booking is processed successfully (online payment) or when the Admin finalizes the check-out.

1. The system compiles the booking information: room, number of nights, services, incidental charges, discount, VAT, total and the amounts already paid.
2. The system issues the invoice (invoice code, issue date, total amount) linked to the booking. If the booking already has an invoice, the system updates the invoice with the latest figures.
3. The system sends a confirmation email containing the invoice information to the customer's email.
4. The system records the result of issuing the invoice.

18.2.2. Alternative flows

18.2.2.1. Failed to send email

If the system cannot send the email, the invoice is still issued, the system logs the error and allows the Admin to resend it later.

18.2.2.2. Booking has no successful payment

If the booking has no successful payment and has not been checked out, the system does not issue an invoice and the use case ends.

18.3. Special requirements

- The invoice must contain full details of all items and absolutely accurate amounts.
- Each booking has at most one invoice.

18.4. Pre-conditions

The booking has a successful payment or is being finalized for check-out by the Admin.

18.5. Post-conditions

If the use case succeeds, the invoice of the booking is issued or updated and a confirmation email is sent to the customer. Otherwise, the system state is unchanged.

18.6. Extension points

None.

19. Use-case View my bookings

19.1. Summary

This use case allows a customer to view their bookings by status.

19.2. Flow of events

19.2.1. Basic flow

The use case begins when the customer selects the function to view their bookings.

1. The system displays the customer's list of bookings in descending order (newer bookings first), including booking code, check-in date, check-out date, total amount and status (Pending, Confirmed, Checked-in, Completed, Cancelled).
2. If the customer chooses to filter by a status, the system displays only the bookings in that status.
3. The customer selects the “View details” function for a booking.
4. The system displays the detailed information of the booking, including: list of rooms, extra services, applied vouchers, deposit, VAT, total amount, payment method, payments made, invoice (if any) and booking status.

19.2.2. Alternative flows

19.2.2.1. Customer has no bookings

If the customer has no bookings (or no bookings in the filtered status), the system notifies that there are no bookings.

19.3. Special requirements

- A customer can only view their own bookings.

19.4. Pre-conditions

The customer (Customer) must be logged in to the system.

19.5. Post-conditions

None.

19.6. Extension points

- Use case Cancel booking: If the customer chooses to cancel a booking (in “Pending” or “Confirmed” status).
- Use case Review stay: If the customer chooses to review a booking in “Completed” status.

20. Use-case Cancel booking

20.1. Summary

This use case allows a customer to cancel a booking according to the hotel's cancellation policy.

20.2. Flow of events

20.2.1. Basic flow

The use case begins when the customer chooses to cancel a booking.

1. The system displays a confirmation dialog for the cancellation, stating the cancellation policy (free cancellation up to 24 hours before check-in time) and the refund amount.
2. If the customer agrees, the system checks that the cancellation is valid (the booking belongs to the customer and its status is Pending or Confirmed).
3. The system changes the booking status to “Cancelled” and releases the held room.
4. If the booking was paid online, the system performs the Refund payment function.
5. The system notifies the customer of the cancellation result.

20.2.2. Alternative flows

20.2.2.1. Customer does not agree to cancel

If the customer does not agree in the confirmation dialog, the system closes the dialog and leaves the booking unchanged.

20.2.2.2. Booking cannot be cancelled

If the booking has been checked in, completed, already cancelled, or does not belong to the customer, the system displays an error message and does not change the booking.

20.2.2.3. Cancellation less than 24 hours before check-in

If the cancellation is made less than 24 hours before the check-in time, the confirmation dialog warns the customer that the deposit will not be refunded. If the customer still agrees, the booking is changed to “Cancelled” without a refund.

20.3. Special requirements

- Free cancellation applies only up to 24 hours before check-in time.
- A customer can only cancel their own bookings.

20.4. Pre-conditions

The customer must be logged in and viewing their list of bookings.

20.5. Post-conditions

If the use case succeeds, the booking is changed to “Cancelled” and the room is released. Otherwise, the system state is unchanged.

20.6. Extension points

- Use case Refund payment: If the booking was paid online and the cancellation is eligible for a refund.

21. Use-case Refund payment

21.1. Summary

This use case refunds the deposit to the customer through the payment gateway (Payment System) when a booking paid online is validly cancelled.

21.2. Flow of events

21.2.1. Basic flow

The use case begins when a booking paid online is cancelled (by the customer in accordance with the policy, or rejected or cancelled by the Admin).

1. The system determines the refund amount according to the policy: the full deposit if the customer cancels at least 24 hours before check-in time or if the Admin rejects or cancels the booking; 0 if the customer cancels late.
2. The system sends a refund request (amount, original transaction code) to the payment gateway (Payment System).
3. The payment gateway processes the request and returns the result to the system.

4. If the refund succeeds, the system records the refunded amount and updates the payment status.
5. The system sends an email notifying the customer of the refund.

21.2.2. Alternative flows

21.2.2.1. Nothing to refund

If the refund amount is 0 (late cancellation) or the booking was not paid online, the system does not send a refund request and the use case ends.

21.2.2.2. Refund failed

If the payment gateway cannot perform the refund, the booking remains in “Cancelled” status, the system records the refund as pending, and notifies the Admin to handle it.

21.3. Special requirements

- The refund amount must be absolutely accurate and must not exceed the amount paid.
- Each payment can be refunded only once.

21.4. Pre-conditions

The booking has been cancelled and has a successful online payment.

21.5. Post-conditions

If the use case succeeds, the refunded amount is recorded for the payment and the customer is notified. Otherwise, the system state is unchanged.

21.6. Extension points

None.

22. Use-case Review stay

22.1. Summary

This use case allows a customer to rate a room with a star rating and write a comment after the stay is completed.

22.2. Flow of events

22.2.1. Basic flow

The use case begins when the customer chooses the function to review a room they have stayed in.

1. The system displays the list of rooms the customer has stayed in (belonging to bookings in “Completed” status) that have not yet been reviewed.

2. The customer selects a room and the system displays a review form containing the star rating (1 to 5) and the comment content.
3. The customer enters the star rating and comment and submits the review.
4. The system validates the review (the customer stayed in the room with a Completed booking, the room has not been reviewed by the customer, the star rating is valid) and saves the review together with the review date.
5. The system notifies the result and displays the review on the room detail page.

22.2.2. Alternative flows

22.2.2.1. No rooms eligible for review

If the customer has no Completed booking or has already reviewed all the rooms they stayed in, the system notifies that there are no rooms to review and the use case ends.

22.2.2.2. Missing star rating

If the customer does not select a star rating, the system displays an error message. The user may return to the beginning of the Basic flow or cancel the operation, in which case the use case ends.

22.3. Special requirements

- Each customer can review each room they stayed in only once.

22.4. Pre-conditions

The customer (Customer) must be logged in to the system.

22.5. Post-conditions

If the use case succeeds, the customer's review is added to the system. Otherwise, the system state is unchanged.

22.6. Extension points

None.

23. Use-case Manage room

23.1. Summary

This use case allows an Admin to manage (view, add, edit, delete) rooms and room types in the system.

23.2. Flow of events

23.2.1. Basic flow

The use case begins when the Admin selects the room management function.

1. The system displays the list of room types (paginated).
2. The Admin selects a room type and the system displays the list of rooms belonging to that type.
3. If the Admin selects “Add” (a room type or a room), the Add room sub-flow is performed.
4. If the Admin selects the “Update” function for a room type or a room, the Update room sub-flow is performed.
5. If the Admin selects the “Delete” function for a room type or a room, the Delete room sub-flow is performed.

23.2.1.1. Add room

1. The system asks the Admin to enter the information: for a room type, the room type name and description; for a room, the room number, room type, price per night, capacity, area, status, description and images.
2. The Admin enters the information and requests to add.
3. The system validates the information (room number is not duplicated, price is greater than 0, images are in a valid format) and adds it to the system.
4. The system notifies the result of the addition.

23.2.1.2. Update room

1. The system displays the information of the room type or room to be updated.
2. The Admin edits the information as desired.
3. The Admin requests to update the information.
4. The system validates the information and updates it in the system.
5. The system notifies the result of the update.

23.2.1.3. Delete room

1. The system asks the Admin to confirm the deletion of the room type or room.
2. If the Admin confirms, the system checks that the room type or room has no active bookings (Pending, Confirmed, Checked-in) and that the room type has no rooms belonging to it. If valid, the system removes the room type or room and its related information from the system.
3. The system notifies the result of the deletion.

23.2.2. Alternative flows

23.2.2.1. Invalid input

If the information entered by the Admin is invalid, the system displays an error message. The Admin may correct the information or cancel the operation, in which case the current flow ends.

23.2.2.2. Room type or room cannot be deleted

If the room type or room has active bookings, or the room type still has rooms belonging to it, the system does not allow deletion and suggests changing the room to an unavailable status (out of operation).

23.3. Special requirements

- Lists must be paginated to improve system performance.
- Only an Admin can use this function.

23.4. Pre-conditions

The Admin must be logged in to the system.

23.5. Post-conditions

If the use case succeeds, the room or room type information is added, updated or deleted in the system. Otherwise, the system state is unchanged.

23.6. Extension points

None.

24. Use-case Manage service

24.1. Summary

This use case allows an Admin to manage (view, add, edit, delete) extra services such as spa and breakfast, and to assign services to rooms.

24.2. Flow of events

24.2.1. Basic flow

The use case begins when the Admin selects the service management function.

1. The system displays the list of services (paginated).
2. If the Admin selects “Add”, the Add service sub-flow is performed.
3. If the Admin selects the “Update” function for a service, the Update service sub-flow is performed.
4. If the Admin selects the “Delete” function for a service, the Delete service sub-flow is performed.
5. If the Admin selects the “Assign rooms” function for a service, the Assign service to rooms sub-flow is performed.

24.2.1.1. Add service

1. The system asks the Admin to enter the service information: service name and service price.

2. The Admin enters the information and requests to add.
3. The system validates the information (name is not duplicated, price is greater than or equal to 0) and adds the service to the system.
4. The system notifies the result of the addition.

24.2.1.2. Update service

1. The system displays the information of the service to be updated.
2. The Admin edits the information as desired.
3. The Admin requests to update the information.
4. The system validates the information and updates it in the system.
5. The system notifies the result of the update.

24.2.1.3. Delete service

1. The system asks the Admin to confirm the deletion of the service.
2. If the Admin confirms, the system checks that the service has not been used in any booking. If valid, the system removes the service and its related information from the system.
3. The system notifies the result of the deletion.

24.2.1.4. Assign service to rooms

1. The system displays the list of rooms, marking the rooms that currently offer the selected service.
2. The Admin selects or deselects the rooms that offer the service and requests to save.
3. The system updates the list of rooms offering the service.
4. The system notifies the result of the update.

24.2.2. Alternative flows

24.2.2.1. Invalid input

If the information entered by the Admin is invalid, the system displays an error message. The Admin may correct the information or cancel the operation, in which case the current flow ends.

24.2.2.2. Service already in use

If the service is already in a booking, the system does not allow deletion and suggests removing the service from the rooms to stop offering it.

24.3. Special requirements

- Changes to a service price apply only to new bookings; existing bookings keep the price at the time of booking.
- Only an Admin can use this function.

24.4. Pre-conditions

The Admin must be logged in to the system.

24.5. Post-conditions

If the use case succeeds, the service information (or the list of rooms offering the service) is added, updated or deleted in the system. Otherwise, the system state is unchanged.

24.6. Extension points

None.

25. Use-case Manage voucher

25.1. Summary

This use case allows an Admin to create, edit and deactivate discount codes (vouchers).

25.2. Flow of events

25.2.1. Basic flow

The use case begins when the Admin selects the voucher management function.

1. The system displays the list of vouchers (paginated) together with their active status.
2. If the Admin selects “Create”, the Create voucher sub-flow is performed.
3. If the Admin selects the “Update” function for a voucher, the Update voucher sub-flow is performed.
4. If the Admin selects the “Deactivate” function for a voucher, the Deactivate voucher sub-flow is performed.

25.2.1.1. Create voucher

1. The system asks the Admin to enter the voucher information: voucher code, discount percentage and expiry date.
2. The Admin enters the information and requests to create.
3. The system validates the information (code is not duplicated, discount percentage from 1% to 100%, expiry date after the current date) and adds the voucher to the system in the active status.
4. The system notifies the result of the creation.

25.2.1.2. Update voucher

1. The system displays the information of the voucher to be updated.
2. The Admin edits the information as desired.
3. The Admin requests to update the information.
4. The system validates the information and updates it in the system.

5. The system notifies the result of the update.

25.2.1.3. Deactivate voucher

1. The system asks the Admin to confirm the deactivation of the voucher.
2. If the Admin confirms, the system changes the voucher to the inactive status; bookings that have already applied this voucher remain unchanged.
3. The system notifies the result.

25.2.2. Alternative flows

25.2.2.1. Invalid input

If the information entered by the Admin is invalid (duplicate code, discount percentage outside the allowed range, invalid expiry date), the system displays an error message. The Admin may correct the information or cancel the operation, in which case the current flow ends.

25.3. Special requirements

- Vouchers are not deleted but only deactivated, to preserve the history of bookings that applied them.
- Only an Admin can use this function.

25.4. Pre-conditions

The Admin must be logged in to the system.

25.5. Post-conditions

If the use case succeeds, the voucher information is created, updated or deactivated in the system. Otherwise, the system state is unchanged.

25.6. Extension points

None.

26. Use-case Manage user

26.1. Summary

This use case allows an Admin to view the list of users, lock or unlock accounts, and create accounts for walk-in guests.

26.2. Flow of events

26.2.1. Basic flow

The use case begins when the Admin selects the user management function.

1. The system displays the list of users (paginated, searchable by email, full name or phone number), including full name, email, phone number, role and account status.

2. If the Admin selects “Add” (walk-in guest), the Create account for walk-in guest sub-flow is performed.
3. If the Admin selects the “Lock” function for an account, the Lock account sub-flow is performed.
4. If the Admin selects the “Unlock” function for an account, the Unlock account sub-flow is performed.

26.2.1.1. Create account for walk-in guest

1. The system asks the Admin to enter the guest's information: full name, email and phone number.
2. The Admin enters the information and requests to create.
3. The system validates the information (email does not already exist), generates a random password, adds the account (role Customer) together with a customer profile to the system, and sends the password to the guest's email.
4. The system notifies the result of the creation.

26.2.1.2. Lock account

1. The system asks the Admin to confirm the account lock.
2. If the Admin confirms, the system locks the account: a locked account cannot log in, and existing bookings are kept unchanged.
3. The system notifies the result.

26.2.1.3. Unlock account

1. The system asks the Admin to confirm the account unlock.
2. If the Admin confirms, the system unlocks the account and resets the failed password attempt count to 0.
3. The system notifies the result.

26.2.2. Alternative flows

26.2.2.1. Invalid input

If the information entered by the Admin is invalid (missing required information, email already exists, invalid format), the system displays an error message. The Admin may correct the information or cancel the operation, in which case the current flow ends.

26.2.2.2. Admin locks their own account

If the Admin chooses to lock their own currently logged-in account, the system does not allow it and displays a message.

26.3. Special requirements

- The user list must be paginated to improve system performance.
- Only an Admin can use this function.

26.4. Pre-conditions

The Admin must be logged in to the system.

26.5. Post-conditions

If the use case succeeds, a new account is added or the lock status of an account is changed in the system. Otherwise, the system state is unchanged.

26.6. Extension points

None.

27. Use-case Manage booking

27.1. Summary

This use case allows an Admin to view all bookings, review them (confirm or reject) and cancel bookings.

27.2. Flow of events

27.2.1. Basic flow

The use case begins when the Admin selects the booking management function.

1. The system displays the list of all bookings in descending order (newer bookings first), paginated and filterable by status.
2. If the Admin selects to view the details of a booking, the system displays the detailed information, including the guest who booked, list of rooms, services, vouchers, incidental charges, payments and booking status.
3. If the Admin selects “Confirm” or “Reject” for a booking in “Pending” status, the Review booking sub-flow is performed.
4. If the Admin selects the “Cancel” function for a booking, the Cancel booking sub-flow is performed.

27.2.1.1. Review booking

1. The system displays the information of the booking in “Pending” status.
2. The Admin selects “Confirm”, or selects “Reject” and enters a reason.
3. If the Admin confirms, the system changes the booking status to “Confirmed”.
4. If the Admin rejects, the system changes the booking status to “Cancelled”, releases the room, and, if the booking was paid online, performs the Refund payment function for the full deposit.
5. The system sends an email notifying the customer of the review result and notifies the Admin of the result.

27.2.1.2. Cancel booking

1. The system asks the Admin to confirm the cancellation and enter a reason.
2. If the Admin confirms, the system checks that the booking is in “Pending” or “Confirmed” status, changes the booking status to “Cancelled” and releases the room.
3. If the booking was paid online, the system performs the Refund payment function for the full deposit.
4. The system sends an email notifying the customer of the cancellation and notifies the Admin of the result.

27.2.2. Alternative flows

27.2.2.1. Booking is not in a suitable status

If the booking has been checked in, completed or cancelled, it cannot be confirmed, rejected or cancelled. The system displays an error message. The Checked-in and Completed statuses change only through the Check-in and Check-out use cases.

27.2.2.2. Refund failed

If the payment gateway cannot perform the refund, the booking is still changed to Cancelled and the refund is recorded as pending for the Admin to handle later.

27.3. Special requirements

- The booking list must be paginated to improve system performance.
- Only an Admin can use this function.

27.4. Pre-conditions

The Admin must be logged in to the system.

27.5. Post-conditions

If the use case succeeds, the booking status is changed (Confirmed or Cancelled) in the system. Otherwise, the system state is unchanged.

27.6. Extension points

- Use case Refund payment: If the booking is rejected or cancelled and was paid online.

28. Use-case Add incidental charge

28.1. Summary

This use case allows an Admin to record incidental charges incurred during a guest's stay on the booking, together with a reason.

28.2. Flow of events

28.2.1. Basic flow

The use case begins when the Admin selects the function to record an incidental charge for a booking of a guest currently staying.

1. The system asks the Admin to select a booking (searching by booking code, room number or guest name) and displays the booking details together with the incidental charges already recorded.
2. The Admin enters the incidental amount and the reason, then requests to record it.
3. The system validates the input (the booking is in “Checked-in” status, the amount is greater than 0, the reason is not empty), adds the incidental charge to the booking and recalculates the total.
4. The system notifies the result of the recording.

28.2.2. Alternative flows

28.2.2.1. Booking is not in a suitable status

If the booking is not in “Checked-in” status, the system displays an error message and does not record the charge.

28.2.2.2. Invalid input

If the amount is invalid or the reason is empty, the system displays an error message. The Admin may correct the information or cancel the operation, in which case the current flow ends.

28.3. Special requirements

- The incidental amount must be added to the total with absolute accuracy.
- Only an Admin can use this function.

28.4. Pre-conditions

The Admin must be logged in to the system.

28.5. Post-conditions

If the use case succeeds, the incidental charge (amount, reason) is recorded on the booking and the booking total is updated. Otherwise, the system state is unchanged.

28.6. Extension points

None.

29. Use-case Check-in

29.1. Summary

This use case allows an Admin to perform the check-in procedure: confirming the guest's arrival and booking code, verifying the citizen ID (if provided), and changing the booking status to staying.

29.2. Flow of events

29.2.1. Basic flow

The use case begins when the Admin selects the check-in function.

1. The system asks the Admin to enter the booking code, or the guest's full name and phone number, to find the booking.
2. The system displays the bookings in “Confirmed” status with a check-in date of today that match the search information.
3. The Admin selects the booking to process and the system displays the booking details, including the room, number of guests, the deposit paid and the remaining amount.
4. The Admin confirms the check-in.
5. The system validates and changes the booking status to “Checked-in” (staying).
6. The system notifies the result of the check-in.

29.2.2. Alternative flows

29.2.2.1. No matching booking found

If no booking matches the search information, the system notifies that none was found. The Admin may search again or end the use case.

29.2.2.2. Booking not eligible for check-in

If the booking is in “Pending” status (not yet confirmed), cancelled or already checked in, the system states the reason and does not allow check-in. If the check-in date has not yet arrived, the system warns that it is not yet the check-in date.

29.3. Special requirements

- The citizen ID number is sensitive personal data; it is protected and can be viewed only by Admin.

29.4. Pre-conditions

The Admin must be logged in to the system.

29.5. Post-conditions

If the use case succeeds, the booking changes to “Checked-in” (staying) status. Otherwise, the system state is unchanged.

29.6. Extension points

None.

30. Use-case Check-out

30.1. Summary

This use case allows an Admin to perform the check-out procedure: finalizing incidental charges, recording the remaining payment, issuing the invoice and marking the booking as completed.

30.2. Flow of events

30.2.1. Basic flow

The use case begins when the Admin selects the check-out function.

1. The system asks the Admin to enter the booking code, room number or guest name to find the booking, and displays the matching bookings in “Checked-in” status.
2. The Admin selects the booking to check out.
3. The system displays the booking details, including the room charge, booked services, vouchers, VAT, recorded incidental charges and the deposit already paid.
4. If the guest has incidental charges not yet recorded, the system performs the Add incidental charge function.
5. The system finalizes the charges, calculates the final total cost and the amount the guest still has to pay (total cost minus the amount already paid).
6. The Admin records the guest's remaining payment (cash or card at the counter) and confirms the check-out.
7. The system performs the Issue invoice function and sends a confirmation email.
8. The system changes the booking status to “Completed” and notifies the result of the check-out.

30.2.2. Alternative flows

30.2.2.1. Booking not found or not checked in

If the booking is not found or is not in “Checked-in” status, the system displays an error message. The Admin may search again or end the use case.

30.2.2.2. Guest has not paid in full

If the recorded amount is less than the amount still to be paid, the system does not allow the check-out to be completed and asks the Admin to record the full payment.

30.3. Special requirements

- The amounts calculated and displayed on the invoice must be absolutely accurate.

30.4. Pre-conditions

The Admin must be logged in to the system.

30.5. Post-conditions

If the use case succeeds, the booking changes to “Completed” status, the remaining payment is recorded, the invoice is issued and the customer can review the stay. Otherwise, the system state is unchanged.

30.6. Extension points

- Use case Add incidental charge: If the Admin needs to record additional incidental charges before finalizing the check-out.

31. Use-case View revenue

31.1. Summary

This use case allows an Admin to view the hotel's revenue over time.

31.2. Flow of events

31.2.1. Basic flow

The use case begins when the Admin selects the view revenue function.

1. The system asks the Admin to select a time range (from date, to date) and a statistics mode (by day or by month).
2. The Admin selects the criteria and requests to view the revenue.
3. The system compiles and displays the revenue as a table and a chart, including the total revenue and the number of completed bookings in each period, as well as the total revenue of the whole range.

31.2.2. Alternative flows

31.2.2.1. Invalid time range

If the start date is after the end date, the system displays an error message. The Admin may select again or cancel the operation, in which case the use case ends.

31.2.2.2. No data

If there are no completed bookings in the selected time range, the system notifies that there is no data.

31.3. Special requirements

- Only an Admin can view revenue.
- Revenue is calculated only from bookings in “Completed” status.

31.4. Pre-conditions

The Admin must be logged in to the system.

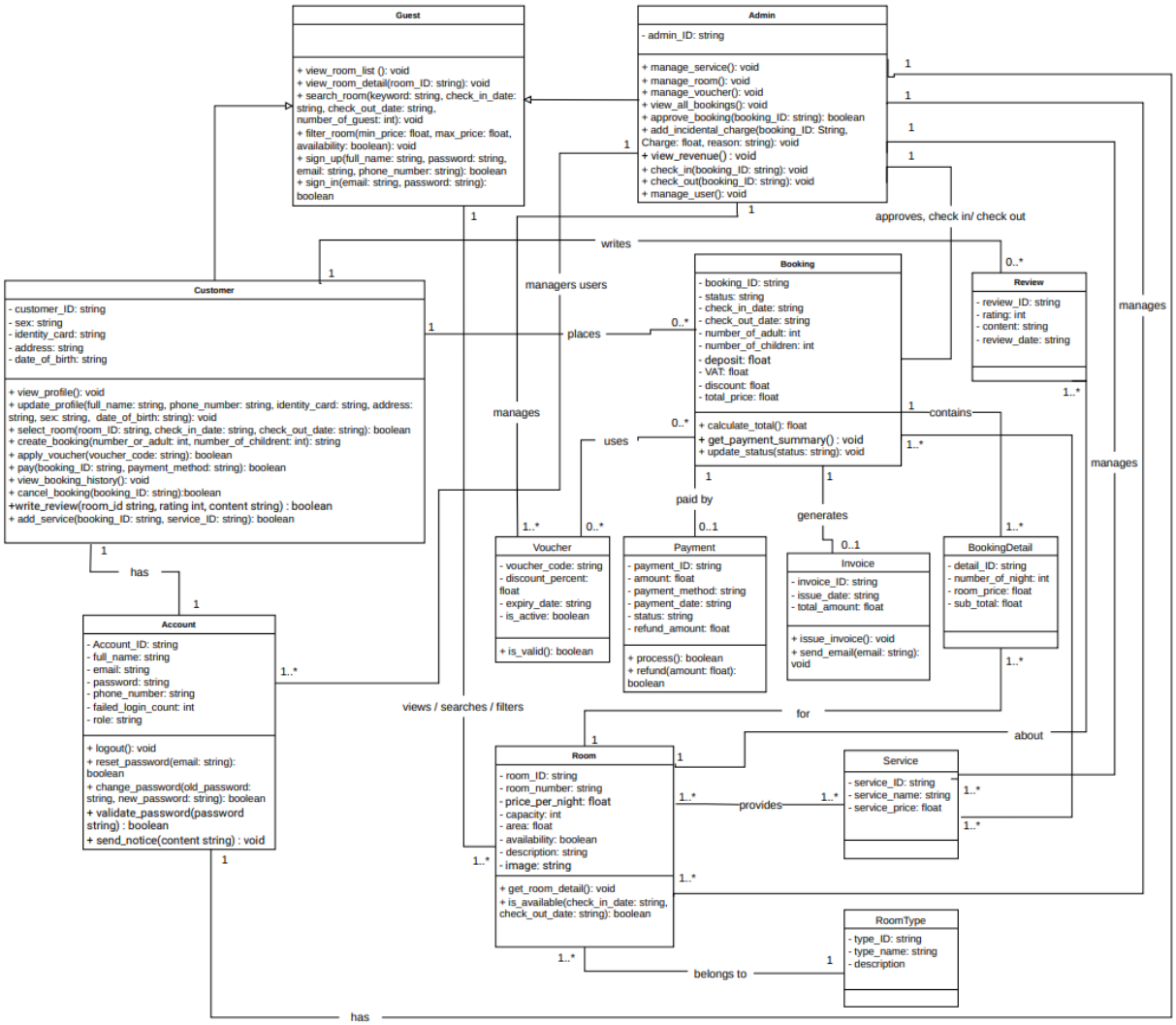
31.5. Post-conditions

None.

31.6. Extension points

None.

V. Class diagram



VI. Interface design and database design.

1. Interface design.

1.1. List of interface pages

No.	Page Name	Description
1	Home page	Main screen; allows searching for rooms by keyword or by selecting check-in and check-out dates.
2	View room	Filters room information by Room ID, check-in date, check-out date, guest capacity, maximum price, minimum price, room availability status, and rating.
3	View room detail	Displays detailed room information, allows selecting check-in/check-out dates, number of adults and children, and adding extra services to make a reservation.
4	My booking	Displays all booked rooms and allows canceling eligible bookings.
5	Booking page	Summarizes the total amount after clicking to book. If the user is not logged in, it automatically redirects to the Sign-in page; if logged in, it redirects to the Payment page.
6	Sign in	Logs users in via email and password with a "Remember Me" option. Includes a redirect link/button to the Sign-up page if the user doesn't have an account, and to the Reset Password page if they forgot their password.
7	Sign up	Allows users to create a new account by entering their email or phone number, password, and full name. Features a show/hide password toggle for convenience. Users confirm registration with the "Sign up" button, or use the "Sign in" button to redirect to the login page if they already have an account.
8	Forgot Password	Allows users to create a new password when they forget their old one by providing their phone number or email address.
9	Payment page	Allows reviewing the invoice and selecting a bank transfer payment method. Upon successful transfer, it navigates to the Payment successful page.
10	Payment successful	Displays a summary of room booking details, sends a confirmation notification via email, and provides options to return to the Home page or view booking history.
11	Cancel booking	Confirms the room to be canceled and agrees to policies; users can either cancel the request or proceed to receive a refund and navigate back to the Home page.
12	Update profile	Accessible from most pages via the Profile section; allows users to fill in additional personal information (then return to Home) or change their password.

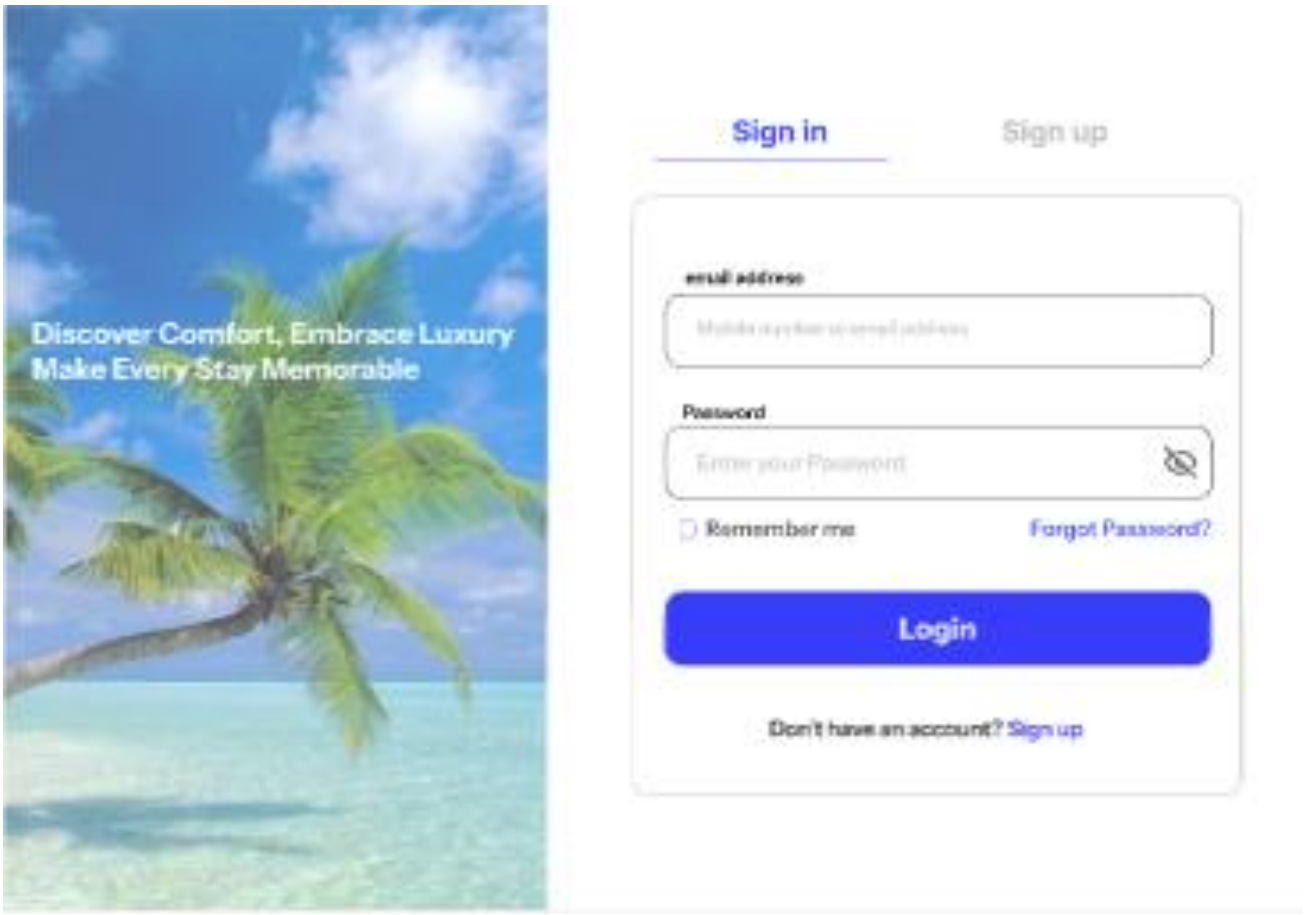
13	Change password	Requires entering the old password, new password, and confirming the new password. Users press Enter to save and return to Home, or click Cancel to return to Home.
14	Review form	Allows users to select star ratings, enter descriptions, upload photos, and view existing reviews. Users can submit or cancel to return to the Home page.
15	Manage booking	Allows staff/admins to check booking statuses and accept/reject reservations: ticking an empty checkbox approves it, an untouched box means pending, and clicking the "X" button rejects it (highlighted in red).
16	Manage user	Displays a user list with User ID, Full Name, National ID/Passport, Address, and Account Status, with capabilities to lock/unlock accounts. Includes navigation to the **Create customer account** page for walk-in guests.
17	Create customer account	Allows staff to fill in details and make a booking for walk-in customers (Full Name, Gender, Phone Number, Address, Date of Birth, Temporary Password, National ID/Passport, and Reserved Room ID).
18	Manage room	Allows adding new room information (Room ID, Room Number, Price per Night, Area, Capacity, Room Type, Description, and Images). Allows viewing updated room details, with options to edit or delete rooms.
19	Revenue report	Displays basic information on booked rooms (Customer ID, Room ID, Total Price, Additional Services, and Status such as payment status, payment method, and room occupancy status). If paid, staff must confirm to verify successful transaction; if unpaid, status can be updated (Checking in, Checked out, or adding additional incurred charges).
20	Manage voucher	Manages voucher details (Voucher Code, Discount Rate/Amount, Start Date, End Date, and Application Conditions). Allows viewing updated vouchers and their current status.
21	Manage service	Manages service information including Service ID, Service Name, and Detailed Description. Displays information for all updated/existing services.

1.2. Interface specification

1.2.1 LOGIN screens

1.2.1.1 LOGIN page

Figure: Sign in



Actor: Guest

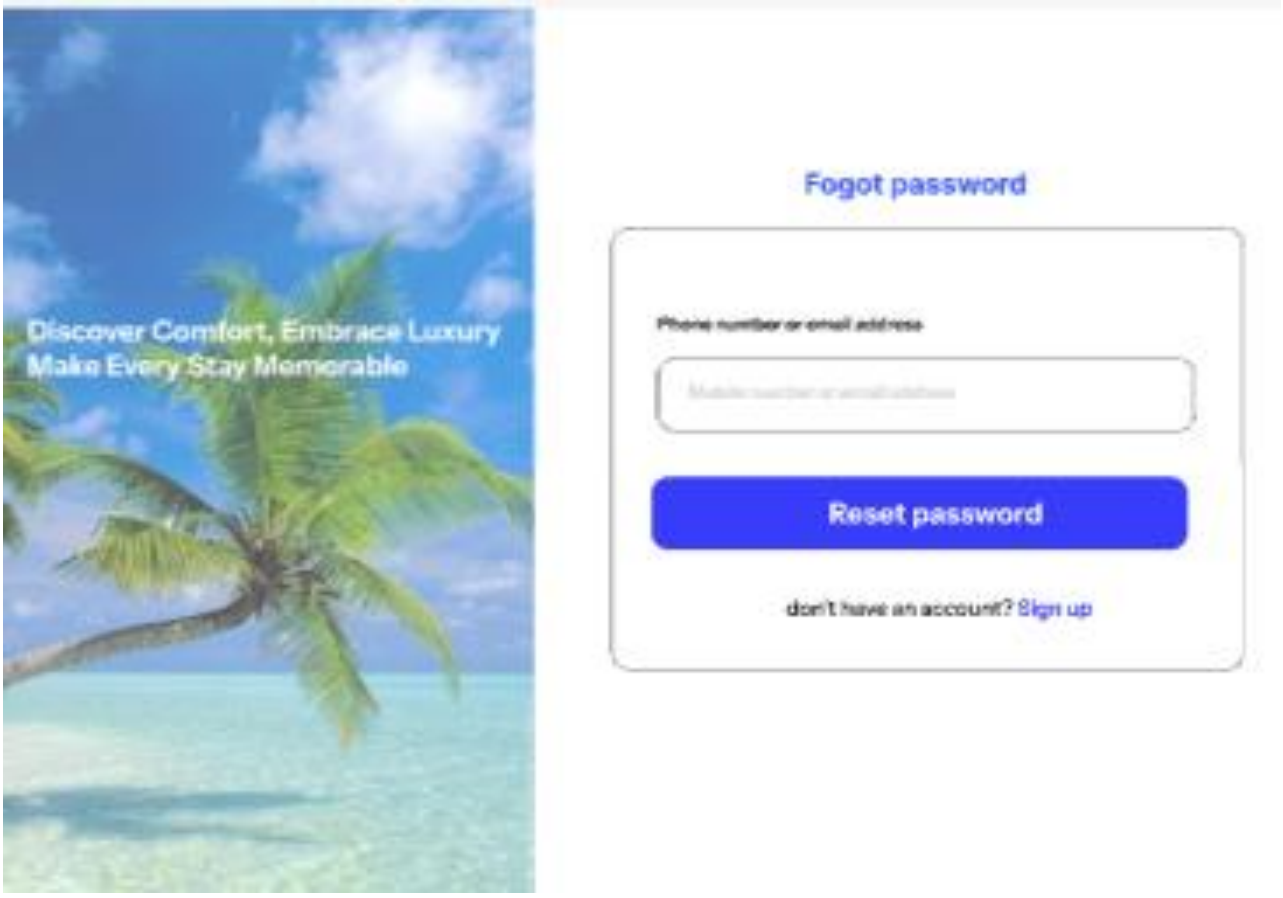
General description: Allows a user to log in with an email and password. After a successful login, the system redirects the user to the page that matches the user's role.

No.	Control	Type	Constraints	Function and handling
1	Email address	Text box	Required, valid email format	Enter the registered email.
2	Password	Text box (masked)	Required	Enter the password.

No.	Control	Type	Constraints	Function and handling
3	Eye icon	Button	None	Show or hide the password.
4	Remember me	Check box	Optional	Keep the login session for longer.
5	Forgot Password?	Link	None	Open the FORGOT PASSWORD page.
6	Login	Button	None	Check the email and password. Correct: a Customer goes to the page they were trying to reach (if any) or the HOME page; an Admin goes to the MANAGE BOOKINGS tab. Incorrect: show an error and record the number of failed attempts. More than 3 failed attempts: temporarily lock login for 15 minutes and send a warning email. Account locked by Admin: tell the user to contact the hotel.
7	Sign up	Link, tab	None	Open the SIGN UP page.

1.2.1.2 FORGOT PASSWORD page

Figure: Forgot password



Actor: Guest

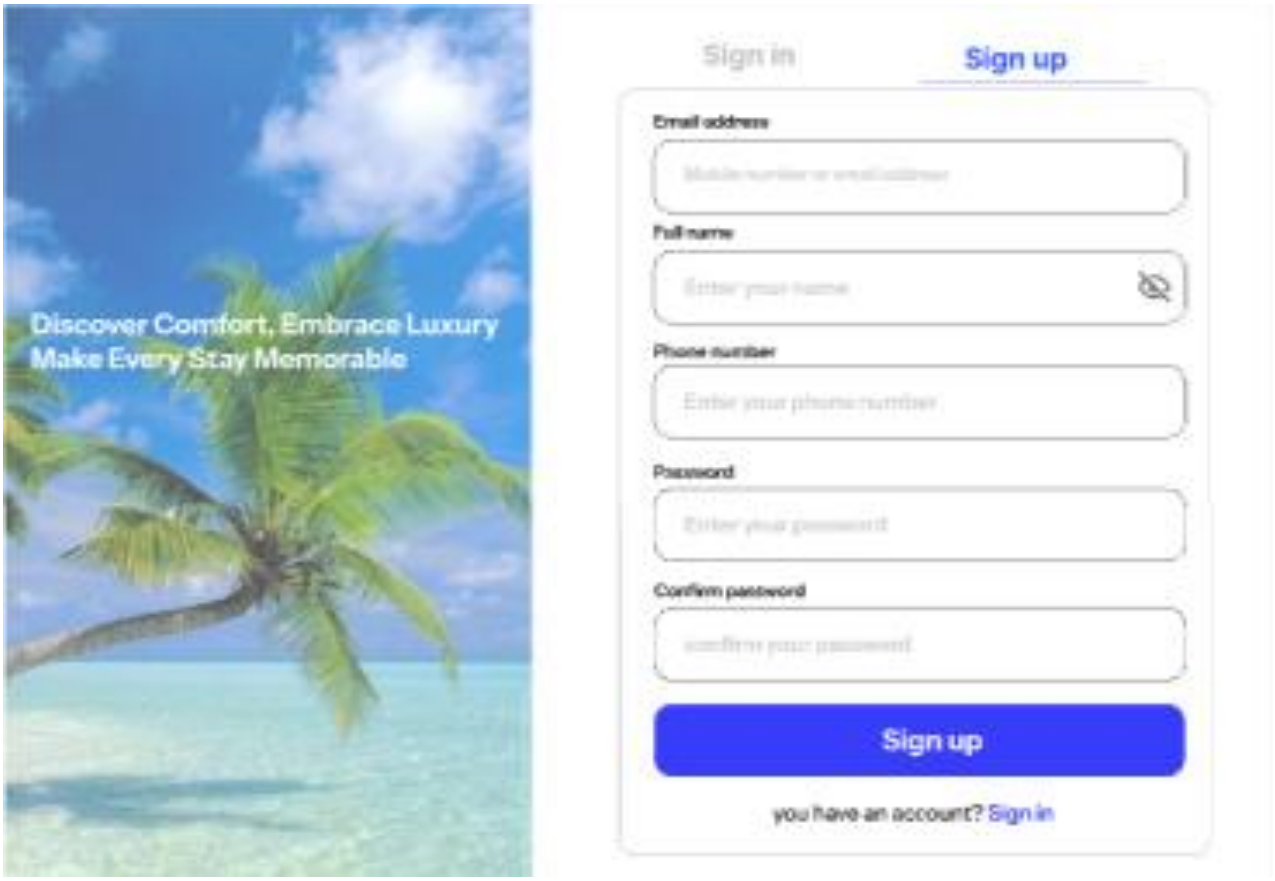
General description: The user enters an email address to receive a new password by email.

No.	Control	Type	Constraints	Function and handling
1	Email address	Text box	Required, valid email format	Enter the registered email.
2	Reset password	Button	None	Email exists: generate a random new password that follows the password rules, send it to the email, and show the result. Email does not exist: show an error. Sending the email fails: show an error and keep the old password.
3	Sign up	Link	None	Open the SIGN UP page.

No.	Control	Type	Constraints	Function and handling
4	Back to sign in	Link	None	Open the LOGIN page.

1.2.1.3 SIGN UP page

Figure: Sign up



Actor: Guest

General description: Creates a new customer account.

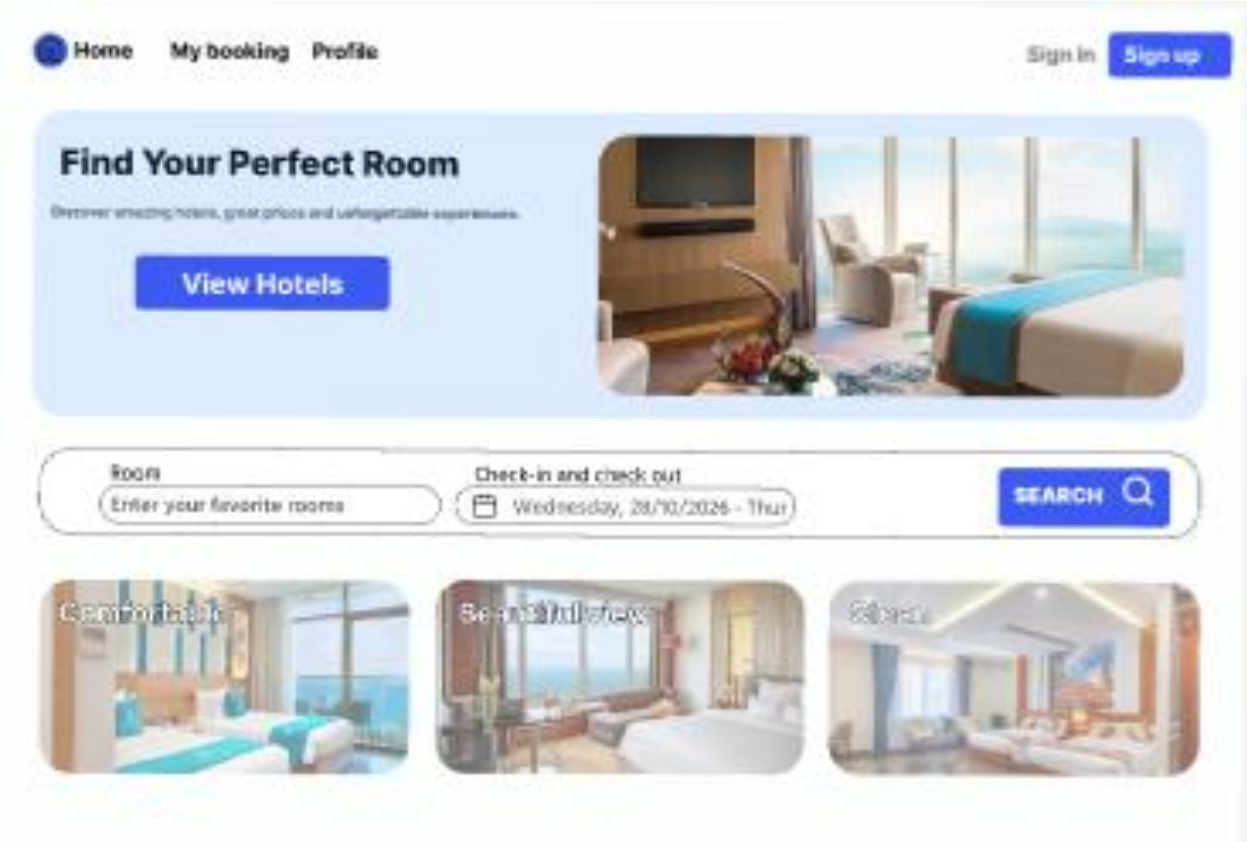
No.	Control	Type	Constraints	Function and handling
1	Full name	Text box	Required	Enter the full name.
2	Email	Text box	Required, valid format, not already registered	Enter the email used to log in.

No.	Control	Type	Constraints	Function and handling
3	Phone number	Text box	Required, valid format, stored as a string	Enter the phone number.
4	Password	Text box (masked)	At least 8 characters with lowercase, uppercase, a digit and a special character	Enter the password.
5	Confirm password	Text box (masked)	Must match Password	Re-enter the password.
6	Sign up	Button	None	Valid: encrypt the password, create an account with the Customer role, show a success message and go to the LOGIN page. Duplicate email or invalid data: show an error that points to the wrong field.
7	Sign in	Link, tab	None	Open the LOGIN page.

1.2.2 Main screens

1.2.2.1 HOME page

Figure: Home



Actor: Guest, Customer, Admin

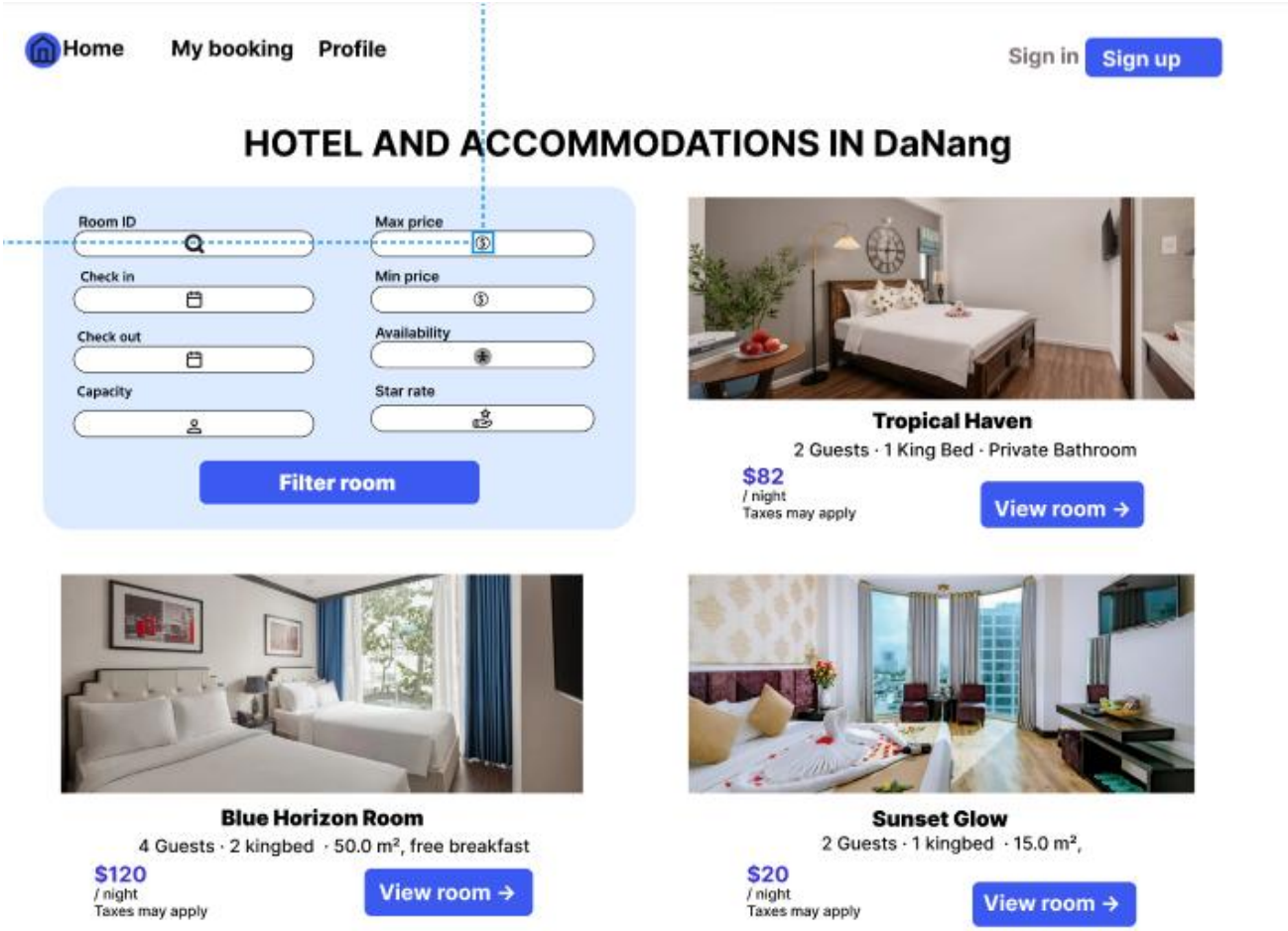
General description: Introduces the hotel and lets users search for rooms quickly.

No.	Control	Type	Constraints	Function and handling
1	Menu (Home, Rooms, Profile)	Link	Profile is shown only after login	Go to the selected page.
2	Sign in, Sign up	Button	Shown to Guest only	Open the LOGIN or SIGN UP page.
3	Logout	Button	Shown after login only	End the session and return to the HOME page.
4	View Rooms	Button	None	Open the ROOM LIST page.

No.	Control	Type	Constraints	Function and handling
5	Room	Text box	Optional	Enter a room name or room ID.
6	Check-in and check-out	Date range picker	Check-in not before today; check-out after check-in	Choose the stay period.
7	Guests	Number box	Greater than 0	Enter the number of guests.
8	SEARCH	Button	None	Validate the criteria. Valid: open the ROOM LIST page with the filtered results. Invalid: show an error.
9	Three photo cards (Comfortable, Beautiful view, Clean)	Clickable card	None	Open the ROOM LIST page for the selected room group.

1.2.2.2 VIEW HOME page

Figure: View home



Actor: Guest, Customer, Admin

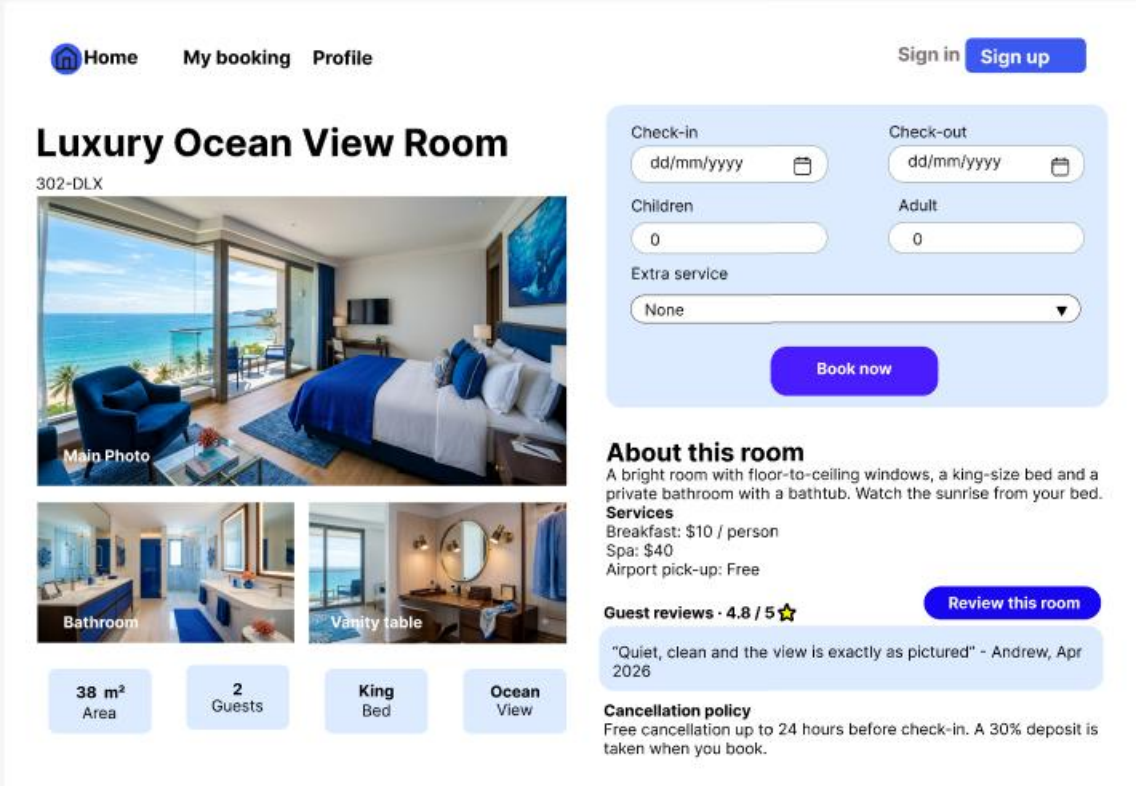
General description: Shows the rooms of the hotel and supports searching, filtering and paging (10 rooms per page). Results are returned within 3 seconds.

No.	Control	Type	Constraints	Function and handling
1	Room ID	Text box	Optional	Search by room ID.
2	Check in, Check out	Date picker	Check-out after check-in	Show only rooms available in the period.
3	Capacity	Drop-down list	Positive integer	Filter by capacity.
4	Min price, Max price	Number box	Not negative; Min ≤ Max	Filter by price range.

No.	Control	Type	Constraints	Function and handling
5	Availability	Drop-down list	None	Filter by availability.
6	Star rate	Drop-down list	None	Filter by average rating.
7	Filter room	Button	None	Reload the list. No match: show “No rooms found” and suggest changing the criteria.
8	Room card (photo, name, capacity, price per night)	Display block	None	Show summary information with a “taxes may apply” note.
9	View room	Button	None	Open the ROOM DETAIL page.
10	Page numbers	Link	None	Go to another page of results.

1.2.2.3 VIEW ROOM DETAIL page

Figure: View room detail



Actor: Guest, Customer, Admin

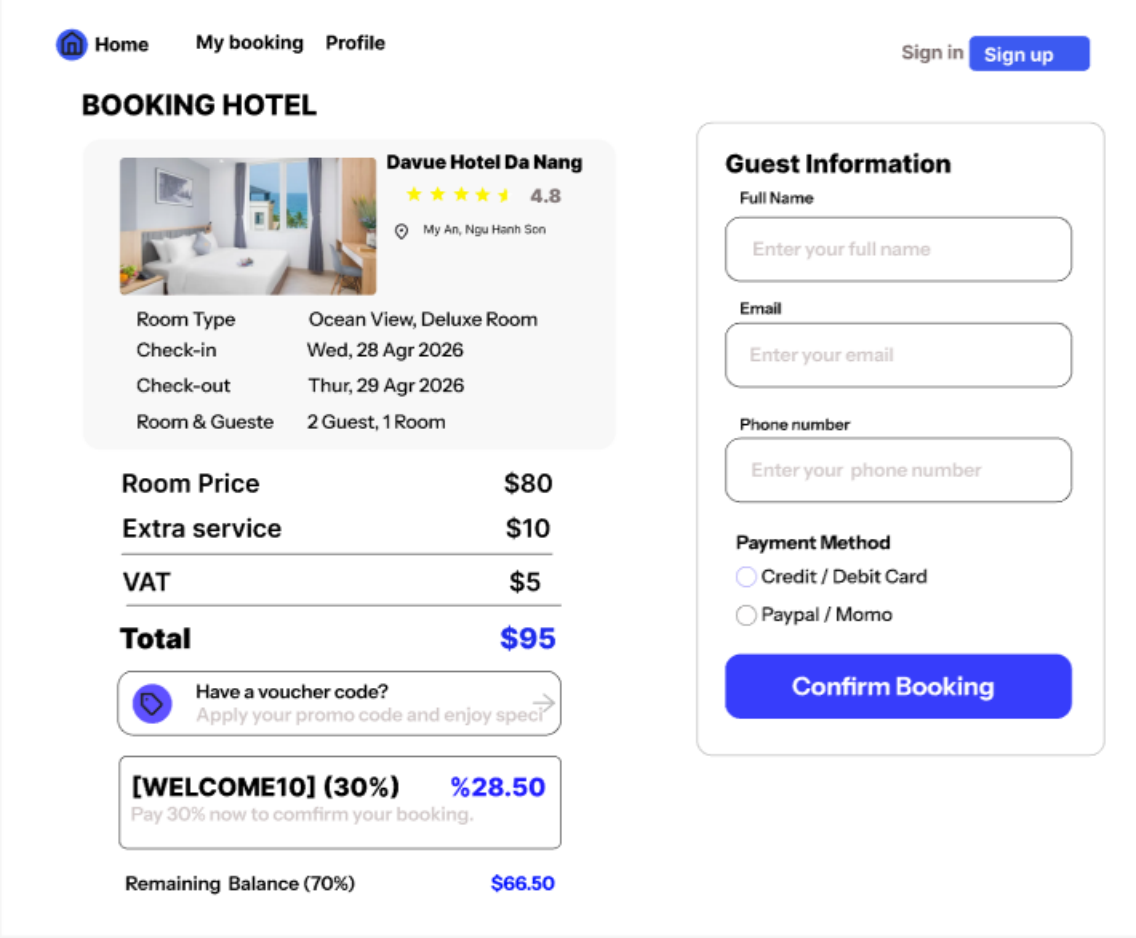
General description: Shows full information about one room and is the starting point for booking. Every user can see rooms and prices.

No.	Control	Type	Constraints	Function and handling
1	Main photo, other photos	Image	None	Click to enlarge.
2	Facts (area, guests, bed, view)	Label	Read-only	Display room facts.
3	About this room, Services	Text block	Read-only	Show the description, included services and prices.
4	Price per night, availability	Label	Read-only	Shown in the booking panel.

No.	Control	Type	Constraints	Function and handling
5	Check-in, Check-out	Date picker	Valid period	Check that the room is available.
6	Children, Adult	Number box	Not negative; total not above capacity	Enter the number of guests.
7	Extra service	Multi-select list	Optional	Choose extra services; their prices are added to the total on the BOOKING page.
8	Book now	Button	None	Guest: go to the LOGIN page and return afterwards. Customer: open the BOOKING page with the selected data. Room not available: show an error.
9	Review this room	Button	Shown only to a Customer with a Completed booking of this room	Open the ROOM REVIEW page.
10	Guest reviews	Display block	Read-only	Show the average rating and recent reviews.
11	Cancellation policy	Text block	Read-only	Free cancellation up to 24 hours before check-in; 30% deposit.

1.2.2.4 BOOKING page

Figure: Booking



Actor: Customer

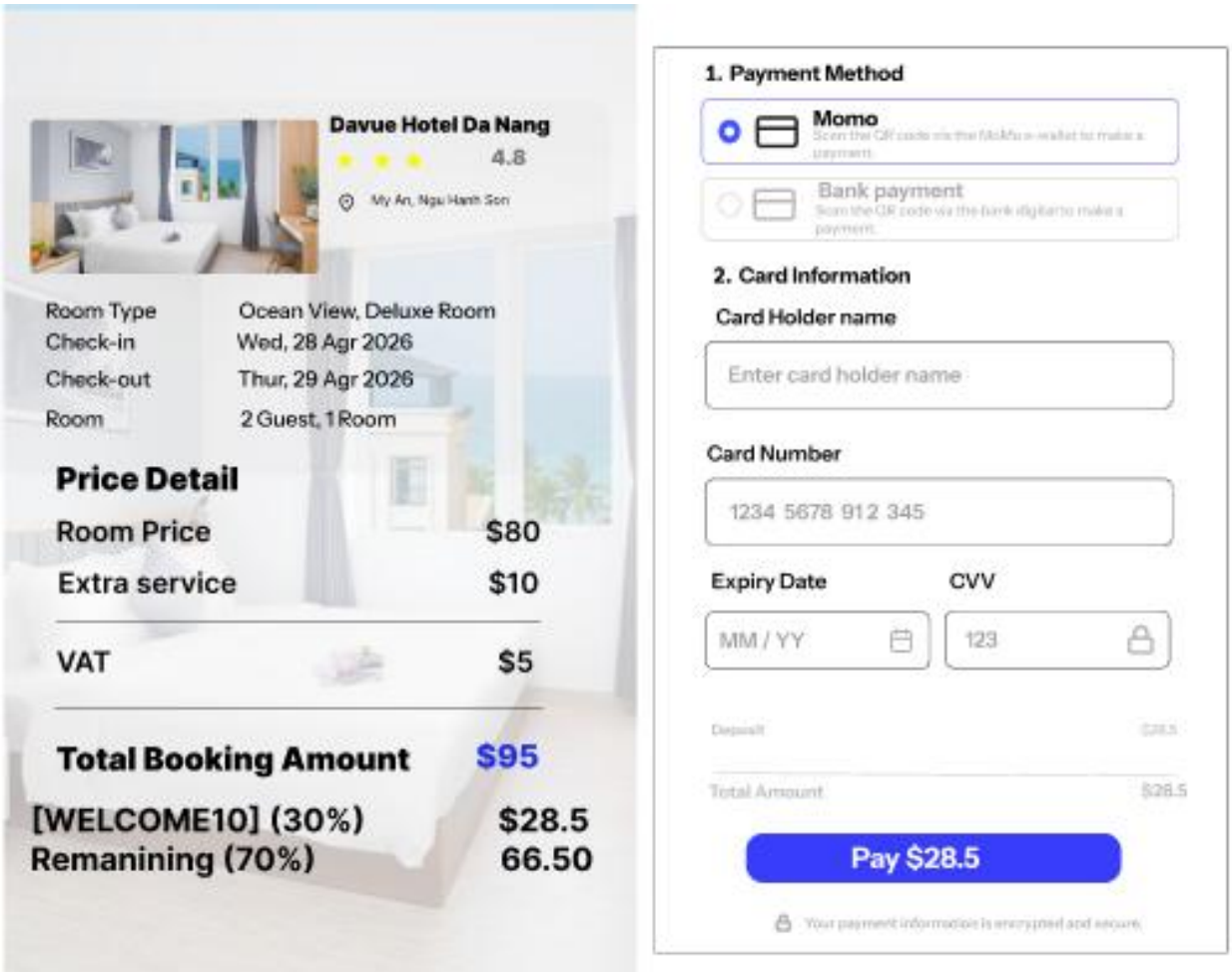
General description: Confirms the booking information, applies a voucher and shows the cost.

No.	Control	Type	Constraints	Function and handling
1	Room summary (hotel, room type, dates, guests)	Label	Read-only	Taken from the ROOM DETAIL page.
2	Room Price, Extra service, VAT, Total	Label	Read-only	Calculated automatically: price × nights + services + tax − discount. Example total: $95.

No.	Control	Type	Constraints	Function and handling
3	Have a voucher code?	Text box with arrow button	Optional	Check the code. Valid: show the discount line and recalculate. Invalid: show the reason (expired, inactive, no uses left).
4	Deposit (30%)	Label	Read-only	30% of the total after discount, for example $28.50.
5	Remaining Balance (70%)	Label	Read-only	The remaining amount paid at the hotel, for example $66.50.
6	Full Name, Email, Phone number	Text box	Required, valid format	Pre-filled from the profile; can be changed if the occupant is not the person booking.
7	Special requests	Text box	Optional	Enter special requests.
8	Confirm Booking	Button	None	Check that the room is still free, hold it for 15 minutes and open the PAYMENT page. Room taken: show a message and return to the ROOM LIST page.

1.2.2.5 PAYMENT page

Figure: Payment



Actor: Customer (the Payment System processes the transaction)

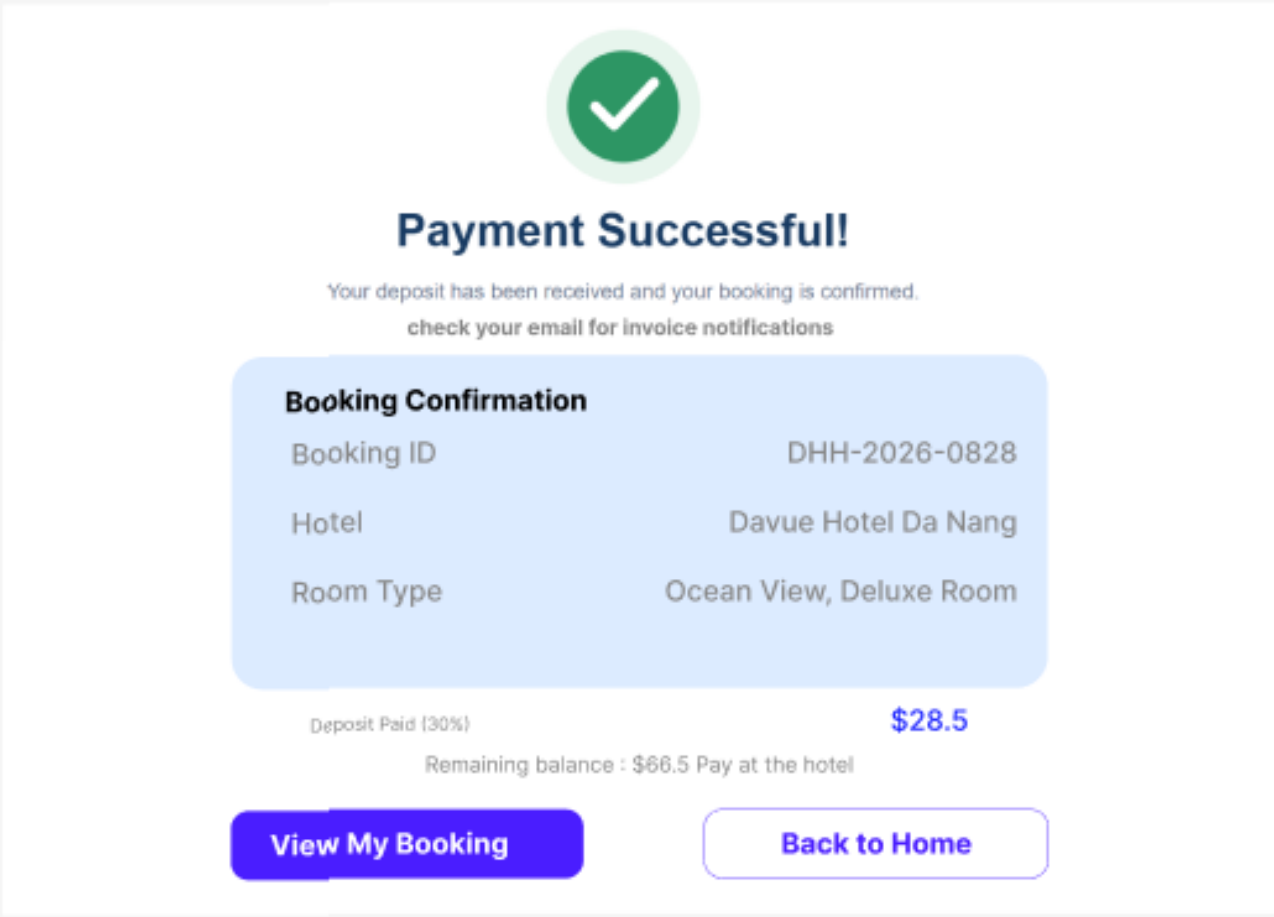
General description: Chooses a payment method and pays the deposit. The system does not store card information.

No.	Control	Type	Constraints	Function and handling
1	Order summary and Price Detail	Label	Read-only	Lets the customer confirm once more. Shows a 15-minute room-hold countdown.
2	Credit / Debit Card	Option button	Choose one	Show the card input section.
3	E-Wallet	Option button	Choose one	Go to the wallet of the Payment System.

No.	Control	Type	Constraints	Function and handling
4	Pay at hotel	Option button	Choose one	Skip the payment gateway, record the booking as Pending and open the PAYMENT SUCCESSFUL page.
5	Card Holder name, Card Number, Expiry Date, CVV	Text box	Required, valid format	Handled by the Payment System; the hotel system does not store them.
6	Pay $28.50	Button	None	Send the deposit to the Payment System. Success: record the deposit and open the PAYMENT SUCCESSFUL page. Failure: show an error and allow retrying, choosing Pay at hotel, or stopping (the room is released). After 15 minutes: release the room and show a time-out message.

1.2.2.6 PAYMENT SUCCESSFUL page

Figure: Payment Successful



Actor: Customer

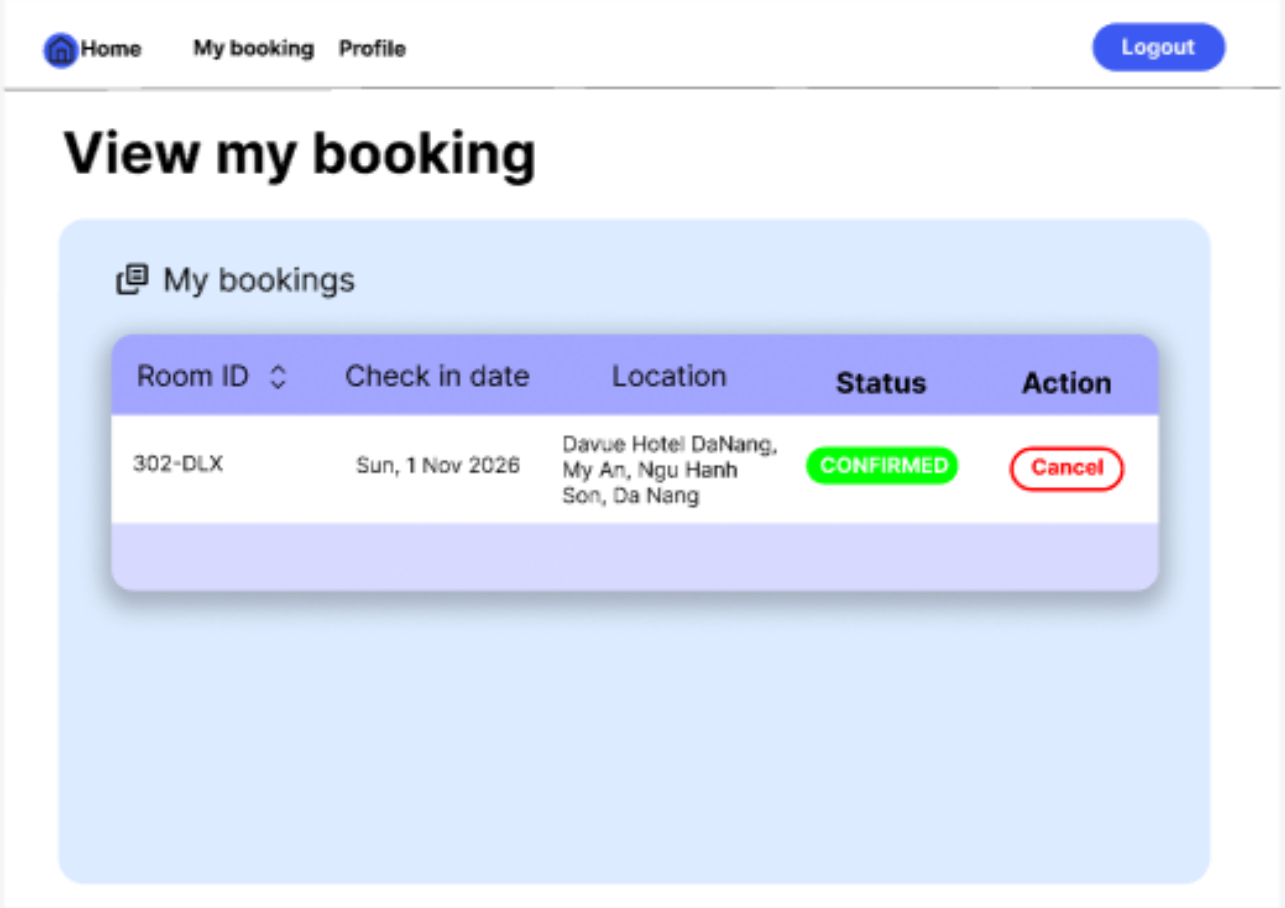
General description: Shows the booking result and confirms that the invoice and confirmation email were sent.

No.	Control	Type	Constraints	Function and handling
1	Booking Confirmation (Booking ID, Hotel, Room Type)	Label	Read-only	Information of the new booking, status Pending.
2	Deposit Paid (30%)	Label	Read-only	$28.50.
3	Remaining balance	Label	Read-only	$66.50, paid at the hotel.

No.	Control	Type	Constraints	Function and handling
4	Email notice	Label	Read-only	The invoice and confirmation email have been sent.
5	View My Booking	Button	None	Open the MY BOOKINGS page.
6	Back to Home	Button	None	Open the HOME page.

1.2.2.7 MY BOOKINGS page

Figure: My booking



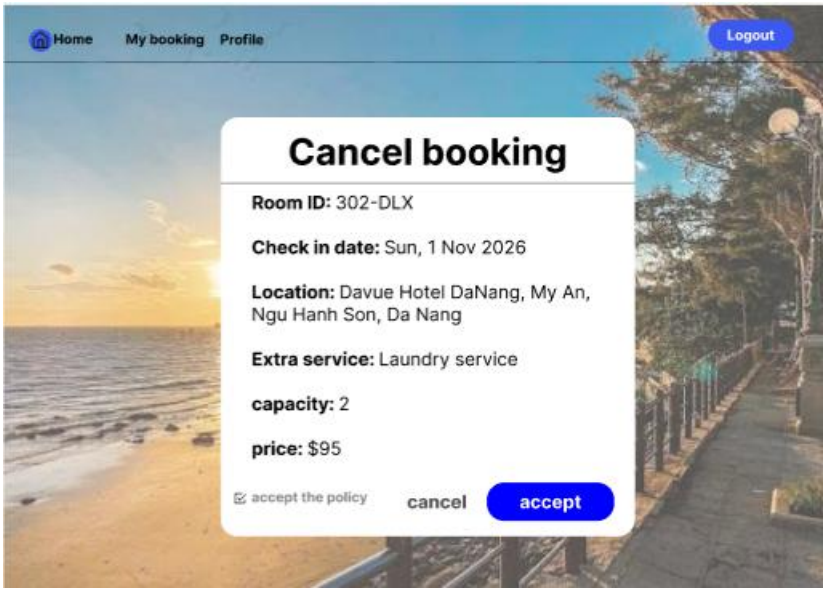
Actor: Customer (sees own bookings only)

General description: Lists the customer's bookings, newest first, with their status and available actions.

No.	Control	Type	Constraints	Function and handling
1	Table (Booking ID, Room ID, Check in, Check out, Location, Total, Status)	Table	Read-only	Show the bookings. No bookings: show a “no bookings yet” message.
2	Sort icon	Button	None	Change the sort order by column.
3	View details	Link	None	See services, voucher, deposit, tax and payments made.
4	Cancel	Button	Shown only for Pending and Confirmed	Open the CANCEL BOOKING window.
5	Review	Button	Shown only for Completed	Open the ROOM REVIEW page.
6	contact support	Button	None	Show the hotel’s contact information.

1.2.2.8 CANCEL BOOKING window

Figure: Cancel booking



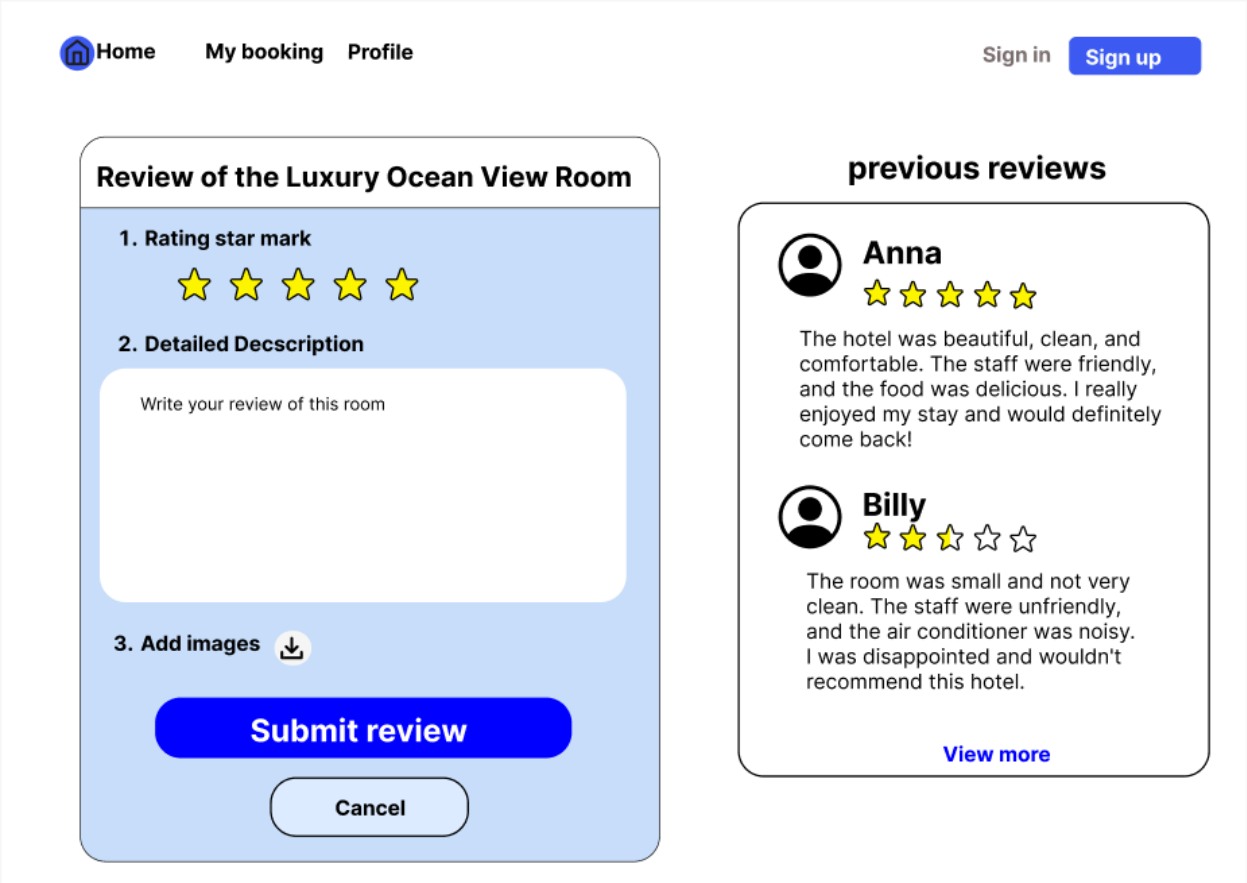
Actor: Customer (the Payment System takes part when a refund is made)

General description: Confirms a cancellation under the hotel's cancellation policy.

No.	Control	Type	Constraints	Function and handling
1	Room ID, Check in date, Location, Extra service, capacity, price	Label	Read-only	Information of the booking to be cancelled.
2	Policy and refund amount	Label	Read-only	Free cancellation up to 24 hours before check-in, for example a $28.50 refund. Less than 24 hours: warn that the deposit is not refunded.
3	accept the policy	Check box	Must be ticked to enable confirmation	The customer agrees to the policy.
4	Keep booking	Button	None	Close the window and keep the booking unchanged.
5	Confirm cancellation	Button	None	Change the booking to Cancelled and release the room. If the deposit was paid online and cancellation is free, send a refund request to the Payment System. Show the result and return to the MY BOOKINGS page.

1.2.2.9 ROOM REVIEW page

Figure: Review



Actor: Customer with a Completed booking of the room

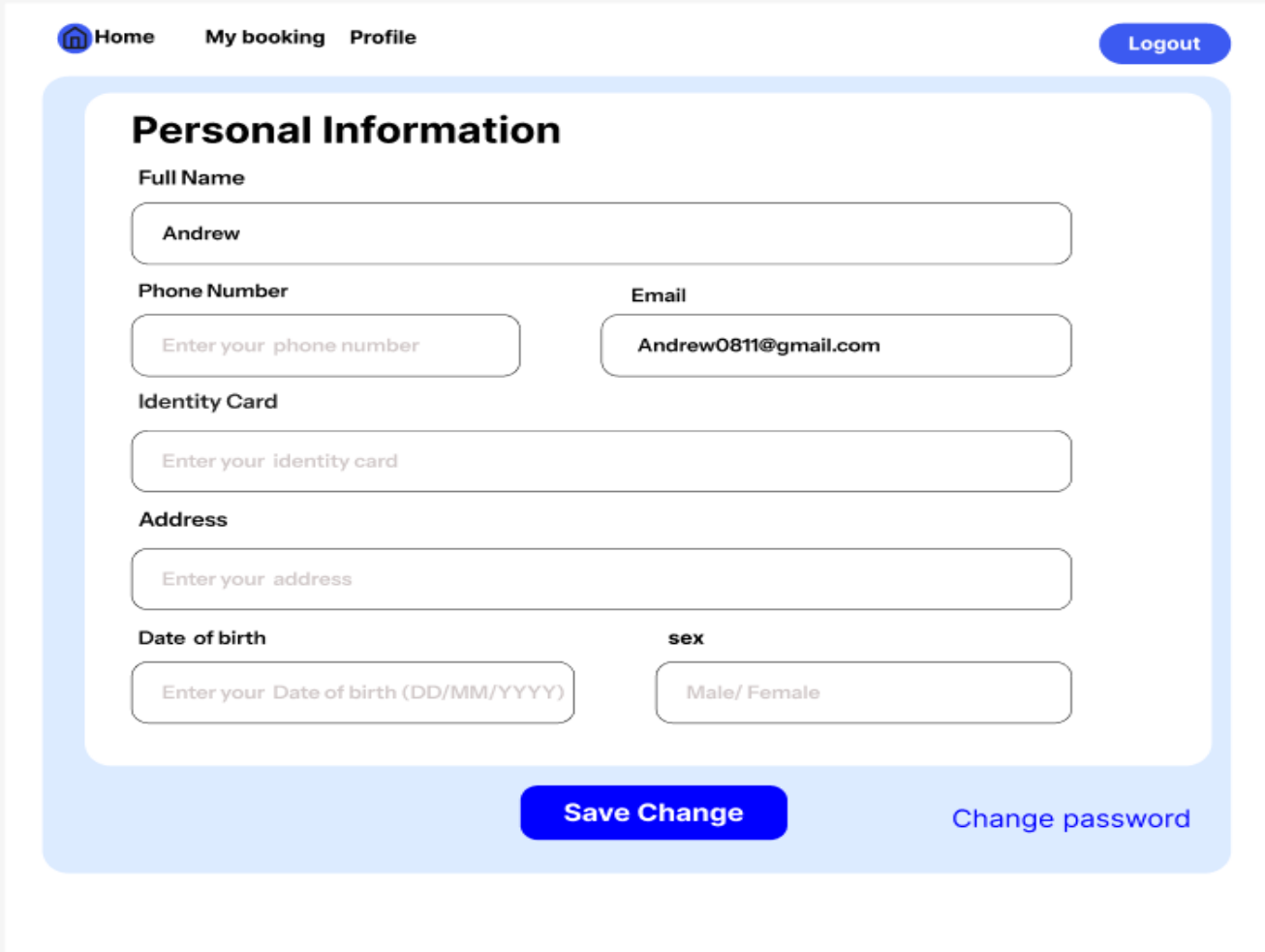
General description: Rates a stay and writes a comment. Each booking can be reviewed only once.

No.	Control	Type	Constraints	Function and handling
1	Rating star mark	5 stars	Integer from 1 to 5	Choose the rating.
2	Detailed Description	Text box	Required	Enter the comment.
3	Submit review	Button	None	Check the conditions (own booking, Completed, not yet reviewed), save the review and show it on the ROOM DETAIL page.
4	Cancel	Button	None	Discard and return to the previous page.

No.	Control	Type	Constraints	Function and handling
5	previous reviews	Display block	Read-only	Reviews from other guests.
6	View more	Link	None	Show more reviews.

1.2.2.10 PROFILE page

Figure: Personal Information



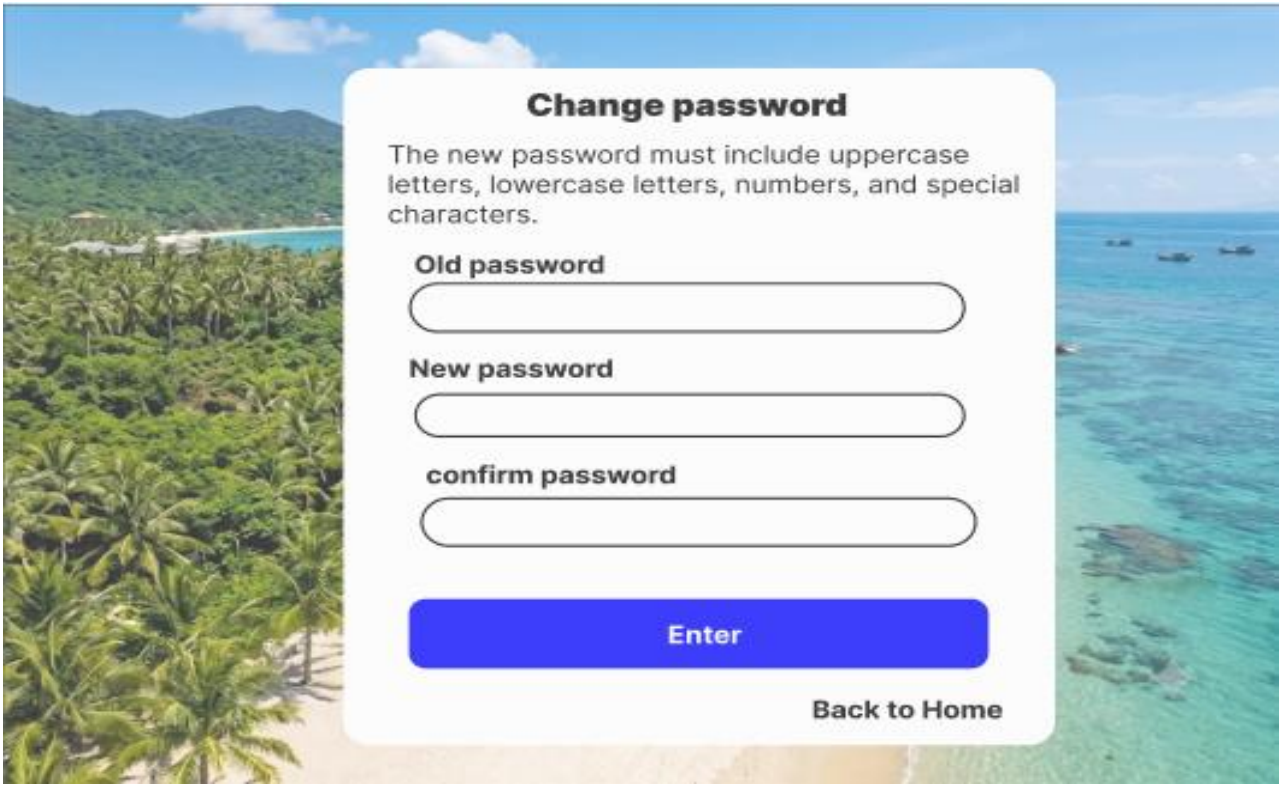
Actor: Customer

General description: Views and updates the customer's own personal information.

No.	Control	Type	Constraints	Function and handling
1	Full Name	Text box	Required	Edit the full name.
2	Phone Number	Text box	Required, valid format	Edit the phone number.
3	Email	Read-only box	Cannot be changed	Display the email.
4	Identity Card	Text box	Stored as a string	Edit the ID card number.
5	Address	Text box	Optional	Edit the address.
6	Date of birth	Date picker	Format DD/MM/YYYY	Edit the date of birth.
7	sex	Drop-down list	Male or Female	Edit the sex.
8	Save Change	Button	None	Validate and save. Invalid data: show an error.
9	Change password	Link	None	Open the CHANGE PASSWORD page.

1.2.2.11 CHANGE PASSWORD page

Figure: Change password



Actor: Customer, Admin

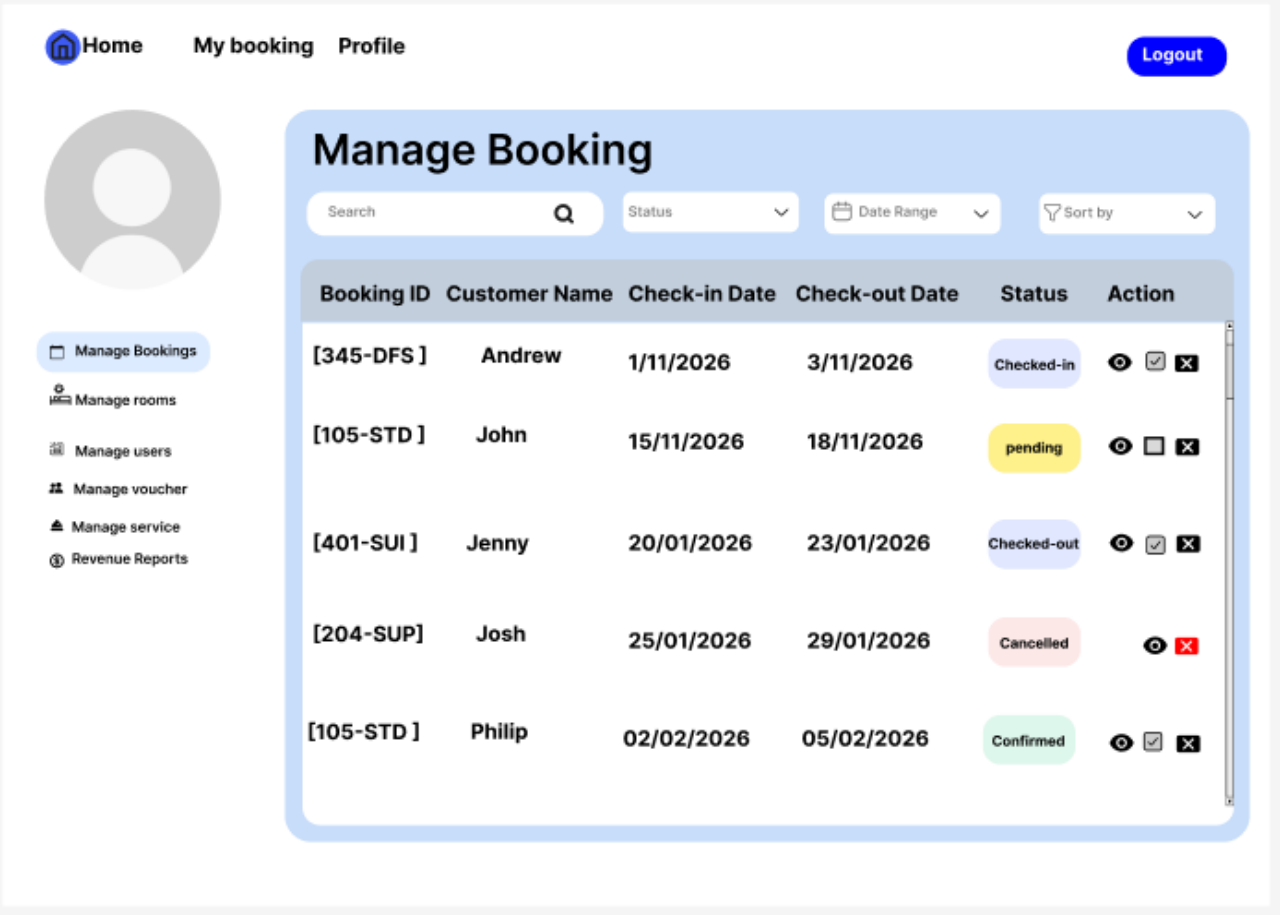
General description: Changes the password; the system sends an email notice when the change succeeds.

No.	Control	Type	Constraints	Function and handling
1	Old password	Text box (masked)	Required, must be correct	Enter the current password.
2	New password	Text box (masked)	At least 8 characters with lowercase, uppercase, a digit and a special character; different from the old password	Enter the new password.
3	confirm password	Text box (masked)	Must match the new password	Re-enter the new password.

No.	Control	Type	Constraints	Function and handling
4	Change password	Button	None	Valid: update the password, send an email notice and show the result. Invalid: show an error that points to the wrong field.
5	Back to Home	Link	None	Open the HOME page.

1.2.2.12 MANAGE BOOKINGS tab

Figure: Manage Booking



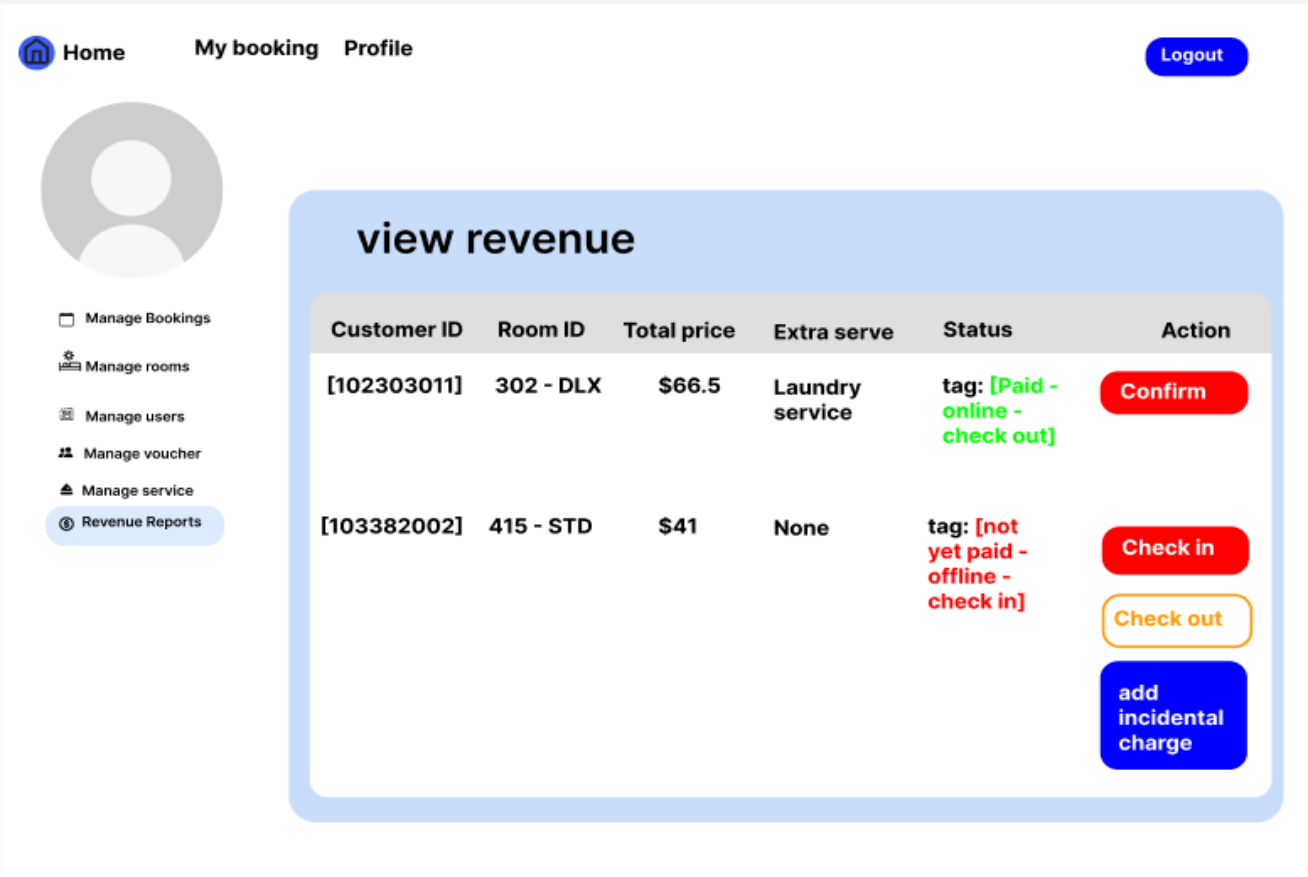
Actor: Admin

General description: Views all bookings, approves or cancels them.

No.	Control	Type	Constraints	Function and handling
1	Search	Search box	Optional	Search by booking ID or customer name.
2	Status	Drop-down list	None	Filter by status.
3	Date Range	Date picker	None	Filter by date range.
4	Sort by	Drop-down list	None	Sort the list.
5	Table (Booking ID, Customer Name, Check-in, Check-out, Status)	Table	Read-only	Show the bookings.
6	Eye icon	Button	None	View booking details.
7	Pencil icon (Approve)	Button	Pending bookings only	Approve the booking and change it to Confirmed.
8	X icon	Button	None	Cancel the booking after confirmation, applying the refund policy.

1.2.2.13 REVENUE REPORT tab

Figure: Revenue report



Actor: Admin

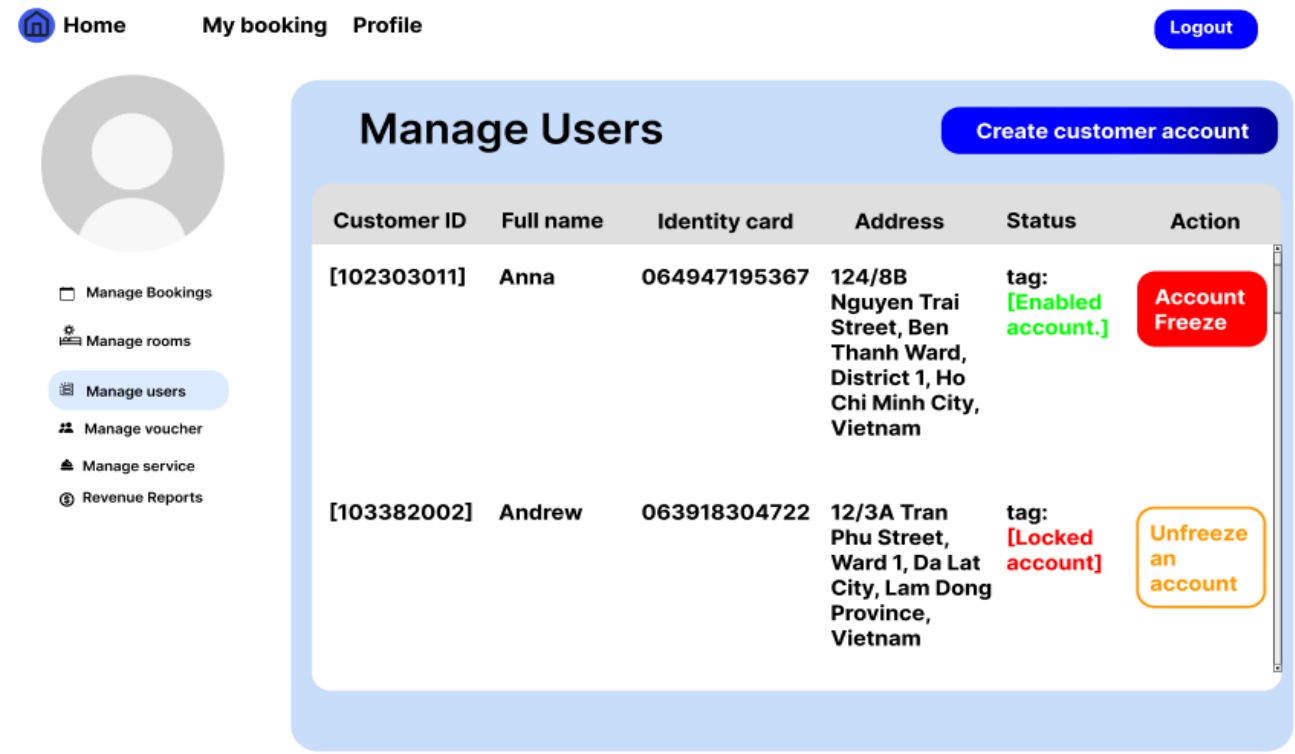
General description: Handles guests arriving and leaving, including walk-in guests.

No.	Control	Type	Constraints	Function and handling
1	Table (Customer ID, Room ID, Total price, Extra service, Status)	Table	Read-only	List of bookings to handle, showing whether payment was made online.
2	Check in	Button	Enabled for Confirmed bookings only	Compare the ID card with the booking information and change the booking to Checked-in.
3	add incidental charge	Button	Checked-in bookings only	Open a form for the amount (not negative) and the reason; save and recalculate the total.

No.	Control	Type	Constraints	Function and handling
4	Check out	Button	Enabled for Checked-in bookings only	Finalize charges, recalculate the total, issue the invoice, change the booking to Completed and free the room.

1.2.2.14 MANAGE USERS tab

Figure: Manage Users



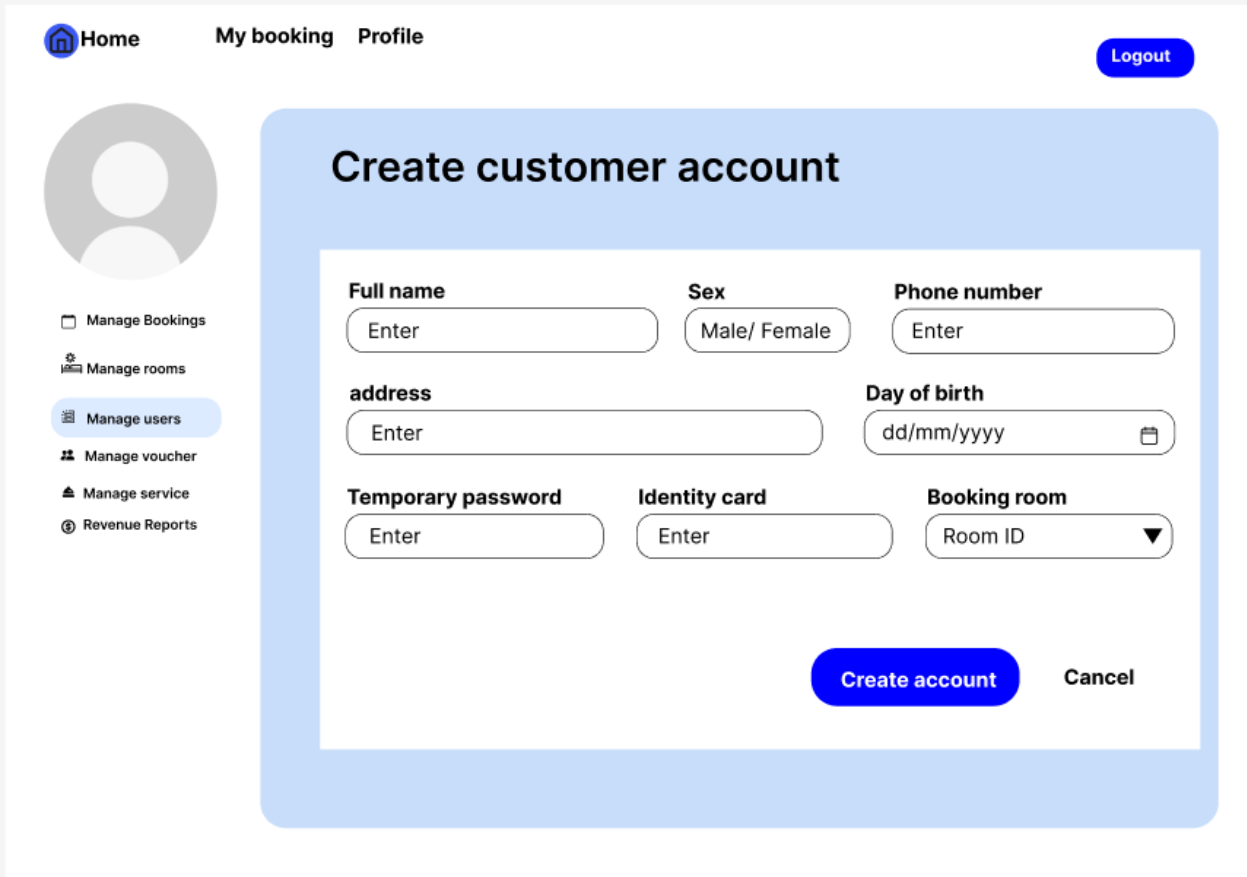
Actor: Admin

General description: Lists customers and locks or unlocks accounts.

No.	Control	Type	Constraints	Function and handling
1	Table (Customer ID, Full name, Identity card, Address, Status)	Table	Read-only	Status is Enabled account or Locked account.
2	Create customer account	Button	None	Open the CREATE CUSTOMER ACCOUNT window.
3	Account Freeze	Button	Shown for Enabled accounts only	Lock the account; the customer can no longer log in.
4	Unfreeze an account	Button	Shown for Locked accounts only	Unlock the account.

1.2.2.15 CREATE CUSTOMER ACCOUNT window

Figure: Create customer account



Actor: Admin

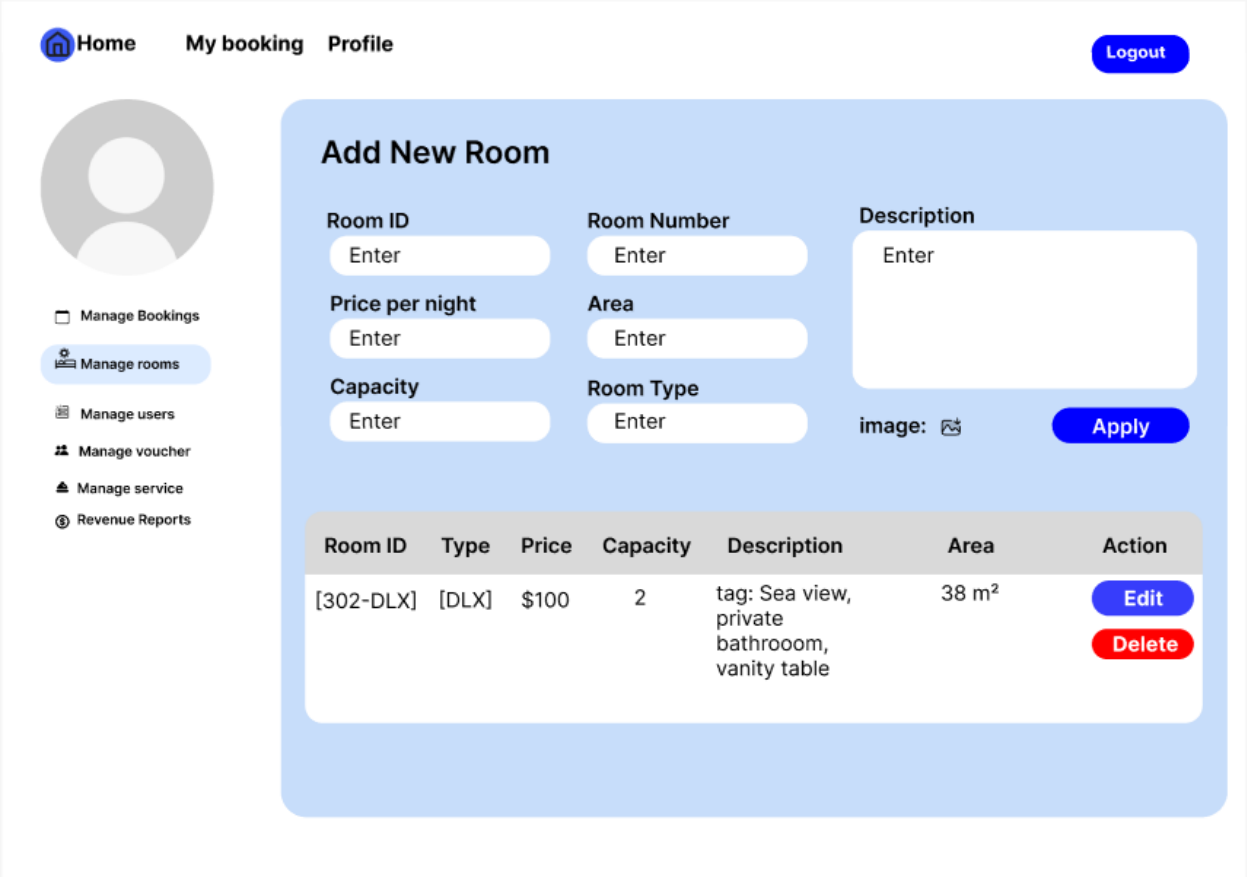
General description: Staff create an account for a walk-in guest and then book a room on the guest's behalf.

No.	Control	Type	Constraints	Function and handling
1	Full name, Sex, Phone number, address, Day of birth, Identity card	Text boxes, drop-down	Full name, phone number and ID card are required	Enter the guest's information.
2	Email	Text box	Required, not already registered	Used to log in and to receive the temporary password.
3	Temporary password	Text box	Follows the password rules	Temporary password sent to the guest by email.

No.	Control	Type	Constraints	Function and handling
4	Booking room	Room ID drop-down	Optional	Choose a room to book for the guest, together with check-in, check-out and number of guests.
5	Create account	Button	None	Valid: create an account with the Customer role and return to the MANAGE USERS tab. Invalid: show an error.
6	Cancel	Link	None	Discard and return to the MANAGE USERS tab.

1.2.2.16 MANAGE ROOMS tab

Figure: Add New Room



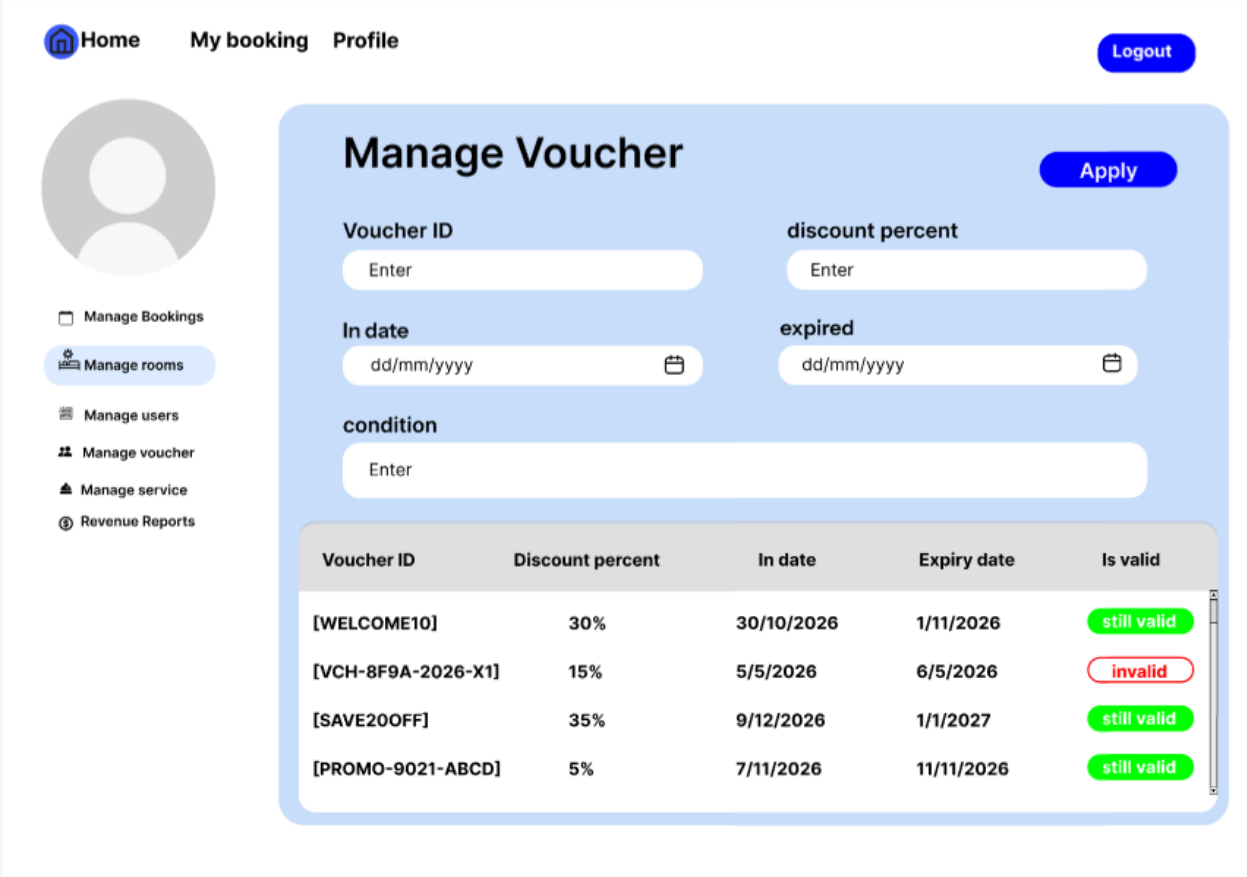
Actor: Admin

General description: Adds, edits and deletes rooms.

No.	Control	Type	Constraints	Function and handling
1	Room ID, Room Number	Text box	Required, unique	Enter the room ID and number.
2	Price per night, Area, Capacity	Number box	Not negative	Enter price, area (m²) and capacity.
3	Room Type	Drop-down list	Required	Choose a room type; a new type can be added.
4	Description	Text box	Optional	Describe the room.
5	image	Upload button	Image formats	Upload room photos.
6	Add room	Button	None	Save the new room and add it to the table.
7	Room table	Table	Read-only	Room ID, Type, Price, Capacity, Description, Area.
8	Edit	Button	None	Load the row into the form, edit and save.
9	Delete	Button	None	Confirm, then delete. A room with active bookings cannot be deleted.

1.2.2.17 MANAGE VOUCHERS tab

Figure: Manage Voucher



Actor: Admin

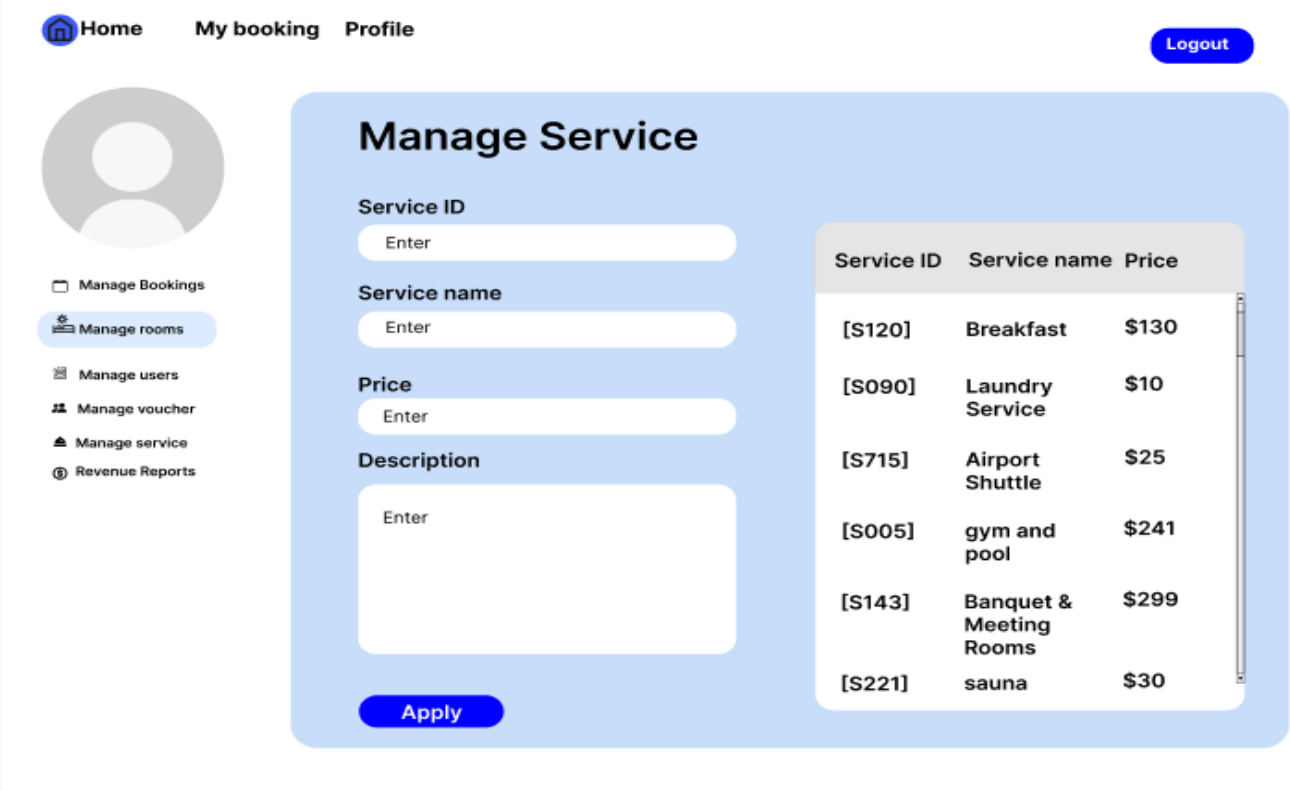
General description: Creates, edits and disables discount codes.

No.	Control	Type	Constraints	Function and handling
1	Voucher ID	Text box	Required, unique	Enter the code.
2	discount percent	Number box	From 1 to 100	Enter the discount percentage.
3	Start date, expired	Date picker	Expiry after start date	Set the validity period.
4	condition	Text box	Optional	Enter the conditions of use.
5	Apply	Button	None	Save a new voucher or update one and add it to the table.

No.	Control	Type	Constraints	Function and handling
6	Table (Voucher ID, Discount percent, Start date, Expiry date, Is valid)	Table	Read-only	Is valid shows still valid, invalid or not started.
7	Edit, Disable	Button	None	Edit or disable the voucher.

1.2.2.18 MANAGE SERVICES tab

Figure: Manage Service



Actor: Admin (design to be added)

General description: Creates, edits and deletes extra services.

No.	Control	Type	Constraints	Function and handling
1	Service ID, Service name	Text box	Required	Enter the service ID and name.
2	Price	Number box	Not negative	Enter the service price.
3	condition	Text box	Optional	Describe the conditions.
4	Apply	Button	None	Save the service.
5	Service table, Edit, Delete	Table, buttons	None	View, edit and delete services. Services are assigned to rooms on the MANAGE ROOMS tab.

2. Database design

